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Physics of Fluids 28, 075102 (2016)
https://doi.org/10.1063/1.4954785

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PHYSICS OF FLUIDS **28**, 075102 (2016)

Numerical and experimental investigation of natural flow-induced vibrations of flexible hydrofoils

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(Received 17 February 2016; accepted 10 June 2016; published online 5 July 2016)

The objective of this work is to present combined numerical and experimental studies of natural flow-induced vibrations of flexible hydrofoils. The focus is on identifying the dependence of the foil's vibration frequencies and damping characteristics on the inflow velocity, angle of attack, and solid-to-fluid added mass ratio. Experimental results are shown for a cantilevered polyacetate (POM) hydrofoil tested in the cavitation tunnel at the French Naval Academy Research Institute (IRENav). The foil is observed to primarily behave as a chordwise rigid body and undergoes spanwise bending and twisting deformations, and the flow is observed to be effectively two-dimensional (2D) because of the strong lift retention at the free tip caused by a small gap with a thickness less than the wall boundary layer. Hence, the viscous fluid-structure interaction (FSI) model is formulated by coupling a 2D unsteady Reynolds-averaged Navier-Stokes (URANS) model with a two degree-of-freedom (2-DOF) model representing the spanwise tip bending and twisting deformations. Good agreements were observed between viscous FSI predictions and experimental measurements of natural flow-induced vibrations in fully turbulent and attached flow conditions. The foil vibrations were found to be dominated by the natural frequencies in absence of large scale vortex shedding due to flow separation. The natural frequencies and fluid damping coefficients were found to vary with velocity, angle of attack, and solid-to-fluid added mass ratio. In addition, the numerical results showed that the in-water to in-air natural frequency ratios decreased rapidly, and the fluid damping coefficients increased rapidly, as the solid-to-fluid added mass ratio decreases. Uncoupled mode (UM) linear potential theory was found to significantly over-predict the fluid damping for cases of lightweight flexible hydrofoils, and this over-prediction increased with higher velocity and lower solid-to-fluid added mass ratio. *Published by AIP Publishing.* [http://dx.doi.org/10.1063/1.4954785]

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I. INTRODUCTION

Previous studies on the flow-induced vibration of flexible foils have mostly focused on aerospace or wind engineering structures, where the fluid density is much smaller than the solid density. However, the solid-to-fluid density ratio is typically between the ranges of 1 and 10 for marine structures. Depending on the direction of motion, the fluid inertial forces can be comparable to or higher than the solid inertial forces for lightweight structures operating in a dense fluid such as water. In addition, the fluid damping and disturbing forces also depend on the relative velocity,[1,2] which suggests that the resulting in-water resonance frequencies will depend on the velocity. The velocity dependence of the natural frequencies and loss factors on the hydrofoils in water has been experimentally or numerically observed by Besch and Liu,[3–5] Blake and Maga,[6] Reese,[7] Seeley

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et al.,[8] Liaghat *et al.*,[9] and Yao *et al.*[10] However, these former studies involved hydrofoils made of relatively heavy and stiff materials (e.g., lead, stainless steel, and aluminum). With an increasing interest in the use of lightweight and composite materials, where the solid-to-fluid density ratio is typically between the ranges of 1 and 2, particularly for high-speed maritime applications, a better understanding of the dynamic response and stability of lightweight hydrodynamic lifting bodies is needed. It should be noted that the operational dependence of the natural frequencies and loss factors has been rarely considered by existing marine designs, as they traditionally use heavy and stiff metallic structures. However, such dependence may be critical for lightweight composite marine structures.

In previous studies of flow-induced vibrations in water, most have focused on cylinders,[11–18] and considered only the heave (or bending) degree-of-freedom (DOF), or coupled heave-surge motion, but not coupled heave-pitch (or bend-twist) motion, as the elastic axis and center of gravity of cylinders coincide, and the center of pressure is close to the elastic axis. For oscillating airfoils, Young and Lai[19] numerically investigated the vortex lock-in phenomenon in the wake of a heaving airfoil; Lua *et al.*[20] and Lua, Lim, and Yeo[21] experimentally studied the wake structures of a heaving or pitching airfoil; and Jung and Park[22] experimentally determined that the vortex shedding frequency varied with the angle of attack and inflow velocity. However, Young and Lai,[19] Lua *et al.*,[20] Lua, Lim, and Yeo,[21] and Jung and Park[22] considered only a simple one degree-of-freedom (1-DOF) airfoil undergoing forced motions at lower Reynolds number ($\mathrm{Re} \leq 2.7 \times 10^4$). The free flow-induced vibrations of a pure pitching airfoil with $4.5 \times 10^4 \leq \mathrm{Re} \leq 1.3 \times 10^5$ was investigated in Poirel, Harris, and Benaissa[23] and Poirel and Yuan;[24] they found that pressure fluctuations generated by the laminar separation bubble and/or unsteady vortex shedding can lead to foil vibrations. The focus of this paper is on the natural flow-induced coupled bend-twist (or heave-pitch) vibrations of cantilevered flexible hydrofoils (where the fluid inertial and damping forces are high compared to structural forces) in fully turbulent flows with $\mathrm{Re} > 3 \times 10^5$. Besides the validations studies referenced or shown in the paper, there are only a few other studies that considered the free bending and twisting vibrations of flexible hydrofoils (e.g., Peng and Zhu[25] and Zhu[26]), and they are mostly limited to lower Reynolds number flows ($\mathrm{Re} \leq 1 \times 10^5$), where viscous effects are more significant and can heavily influence the vortex shedding frequency and foil vibration. Only limited numerical and experimental studies are available for flexible hydrofoils in moderate to high Reynolds number flows, particularly for free vibration studies. Since full-scale hydrodynamic lifting bodies (e.g., hydrofoils, propellers, turbines, rudders, control fins, etc.) typically operate at $\mathrm{Re} = 1 \times 10^5$ or higher, additional studies are needed to understand flow-induced vibrations, resonance frequencies, and total damping coefficients of such devices at various velocities and angles of attack for moderate to high Reynolds number flows.

The objective of this study is to perform combined numerical and experimental studies to investigate the influence of varying velocity, angle of attack, and solid-to-fluid added mass ratio on the natural flow-induced vibrations of hydrofoils, with a special focus on vibration frequencies, total loss factors, elastic deformations, and hydrodynamic load coefficients.

II. EXPERIMENTAL MODEL

Experimental measurements were carried out in a circulating water tunnel at the French Naval Academy Research Institute (IRENav). The test section was 1 m long and had a 0.192 m × 0.192 m square section, as shown in Fig. 1(a). The facility provided regulated velocities ranging from 2 m/s to 12 m/s, and regulated pressures from 30 mbar to 3 bars. The flow rate was regulated with a programmable controller. Two high-precision Paroscientific Series 1000 transmitters were used to measure the static pressure difference between the points of the test section and at the large section before the flow contracts into the test section, as shown in Fig. 1(a). The transmitters have an accuracy of 0.01% for up to 40 psi (2.75 bars). The static pressure differences were then related to the flow rate by using the classical Bernoulli formulation for pipe flow. The mean inflow velocity was measured using a 2D Laser Doppler Velocimetry (LDV) at the inlet of the test section. Taking into account the tunnel regulation system, the uncertainties of the flow rate computed from pressure measurements were ±0.02 m/s.

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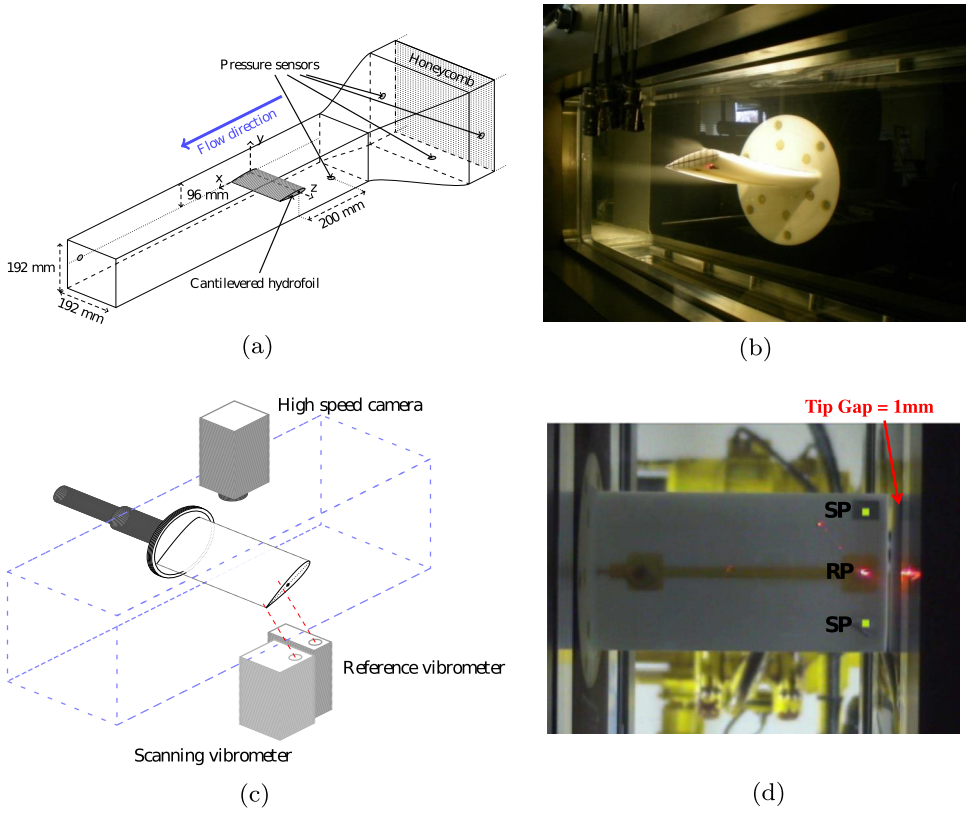


FIG. 1. The French Naval Academy Research Institute (IRENav) experimental setup: (a) cantilevered hydrofoil mounted horizontally inside the water tunnel, (b) sample flow-induced coupled spanwise tip bending and twisting deformations of the NACA0015 POM hydrofoil inside the tunnel, (c) systematic of the foil mounting and Laser Doppler Velocimetry (LDV) setup, and (d) pressure side (bottom) view of the cantilevered NACA0015 polyacetate (POM) hydrofoil inside the test section: the hydrofoil is fixed on the left end, and free on the right end, where there is a 1 mm gap between the foil tip and the tunnel wall; flow goes from bottom to top. Also shown are the scanned measurement points (SP) and fixed reference point (RP) near the free end of the hydrofoil.

The tunnel turbulence intensity was 2%, which is similar to other low turbulence hydrodynamic tunnels and cavitation tunnels.[27–30] In particular, experimental and numerical studies of transition effects[30] in the same cavitation tunnel showed that the 2% inlet turbulence level has very low impact on hydrofoil performances. The tunnel turbulence intensity was measured by LDV at the center of the test-section and 0.3 m upstream of the foil's leading-edge shown in Fig. 1(a).

A flexible, cantilevered, rectangular hydrofoil with a NACA0015 section geometry with a chord of $c = 2b = 0.1$ m and a span of $s = 0.191$ m was used (i.e., a span-to-chord ratio of 1.91). The flexible hydrofoil was made of a polyacetate (POM) with a Young's modulus of $E_s = 2.9$ GPa, Poisson's ratio of $\nu_s = 0.35$, and solid density of $\rho_s = 1420$ kg/m^3. As shown in Fig. 1, the foil was mounted horizontally in the tunnel test section. The foil was clamped at its root into a steel cylinder shaft and was connected to an electric drive to set its undeformed angle of attack (α_o) to a prescribed value. The rotation center, or locus of the shear center, and also called the elastic axis (EA) was at $0.36c$ downstream of the POM hydrofoil's leading-edge. It should be noted that there is a small gap of 1 mm between the free end of the hydrofoil foil and the side wall of the tunnel, as shown in Fig. 1(d). The gap size was kept to a minimum (less than the wall boundary layer thickness at a given velocity, e.g., 9 mm for an inflow velocity of 5 m/s according to Ducoin, Astolfi, and Sigrist[29]) to yield uniform spanwise hydrodynamic loading over majority (~80%) of the hydrofoil. A detailed investigation of the influence of the gap size on the hydrodynamic loading can be found in Harwood and Young.[31] Note that the all IRENav experimental studies reported in this paper were limited to angles of attack less than or equal to 17° to avoid blockage effects caused by the tunnel wall and to avoid excessive loads on the foil.

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To obtain reference rigid hydrodynamic loads for the same foil geometry, a stainless steel NACA0015 hydrofoil (Young's modulus of $E_s = 203$ GPa, Poisson's ratio of $\nu_s = 0.3$, and solid density of $\rho_s = 8010$ kg/m^3) was used. Static hydrodynamic forces were not measured for the flexible POM hydrofoil to avoid modifying the foil response by the mounting setup for the hydrodynamic balance. The steel hydrofoil was mounted in a 3-component hydrodynamic balance measuring the lift, drag, and moment. The axis of rotation for the steel hydrofoil was located at $0.25c$ from the leading-edge. To ensure fully turbulent flow over the steel foil surface, a roughness strip was placed close to the foil leading-edge. Due to the precision of the mounting system, the uncertainty in the initial angle of attack was about ±0.15°. For the hydrodynamic balance, each acquisition lasted 10 s with a sampling frequency of 1 kHz. The experimental accuracy of the balance obtained from calibrations was ±5 N for the lift, ±1.6 N for the drag, and ±1 N m for the moment. The corresponding uncertainties of the measurements are $C_L = \pm 0.02$, $C_D = \pm 0.004$, and $C_M = \pm 0.007$, as shown later in Fig. 10.

The flow-induced vibration measurements were only conducted for the flexible POM hydrofoil. The hydrofoil vibrations were measured by using two LDVs, which are precision optical transducers. The LDV measures the velocity fluctuations at a given point or over a prescribed scanning grid, and its technology is based on the Doppler effect, sensing the frequency shift of back-scattered light from a moving surface. The spatial resolution of the reflected laser beam was less than 1 mm, which is very small compared to the hydrofoil size. Figure 1(c) shows two Polytec® LDVs: PDV-100 for reference and PSV-400 for scanning, using a class II HeNe laser with a wavelength of 633 nm. The first laser measured the vibration velocity of a reference point (RP), which was set at the mid-chord near the free tip of the hydrofoil, as shown by the red dot in Fig. 1(d). The second laser measures the velocity fluctuations at a given scanning point (SP) or over a prescribed scanning grid, as demonstrated via the green dots in Figs. 1(d) and 2(b), respectively. The scanned measurement and fixed reference points were covered with reflective patches to avoid any light-absorption effects due to the hydrofoil material and to improve the signal to noise ratio.

For the vibration mode shape measurements in still air and in still water, an electrodynamic shaker was used to prescribe known input force functions, as shown in Fig. 2(a). Note that the still air and still water mode shape measurements use the forced motion input and hence are different from the free flow-induced vibration measurements. The scanning vibrometer was equipped with motor driven mirrors to allow measurements of the vibration levels on a predefined grid on the structure shown in Figs. 2(b)-2(f). The grid had 9×21 points, with 9 points along the chordwise direction and 21 points along the spanwise direction. This arrangement yielded the same grid spacing in the spanwise and chordwise directions and has shown to be sufficient for the determination of the first bending and twisting modes, as illustrated in Figs. 2(b)-2(f). The phase information throughout the scan was obtained by computing the average cross-spectrum between the reference point and the scanned points of the grid. The mode shapes in dry and in still water conditions were obtained by taking the relative magnitude and phase difference between the scanning points and the reference point, and by applying known and repeatable forced input via the electrodynamic shaker. It should be pointed out that for each mode shape, 16 vibration spectrum were recorded, and the values were averaged at each grid point; the recording time of the vibration measurements for each experimental configuration took up to four hours depending on the frequency resolution.

As shown in Figs. 2(c)-2(f), the first mode of the NACA0015 POM foil corresponds to primarily bending, the second mode corresponds to primarily twisting, and the mode shapes in-air and in-still-water are very similar. Only the first bending and first twisting modes are considered in the numerical model, as they dominate the flow-induced vibration response of the foil, as will be shown later. The measured first three modal frequencies in-air and in-still-water of the cantilevered NACA0015 POM hydrofoil via impulse excitations using an electrodynamic shaker are shown in Fig. 3 and listed in Table I. f_h, f_θ, and $f_{h,2}$ are the first in-air natural bending frequency, the first in-air natural twisting frequency, and the second in-air natural bending frequency, respectively; f_h^*, f_θ^*, and $f_{h,2}^*$ are the corresponding in-water natural frequencies. It should be noted that during the experiments, small disparities occurred on the measured frequencies due to the repeatability of the

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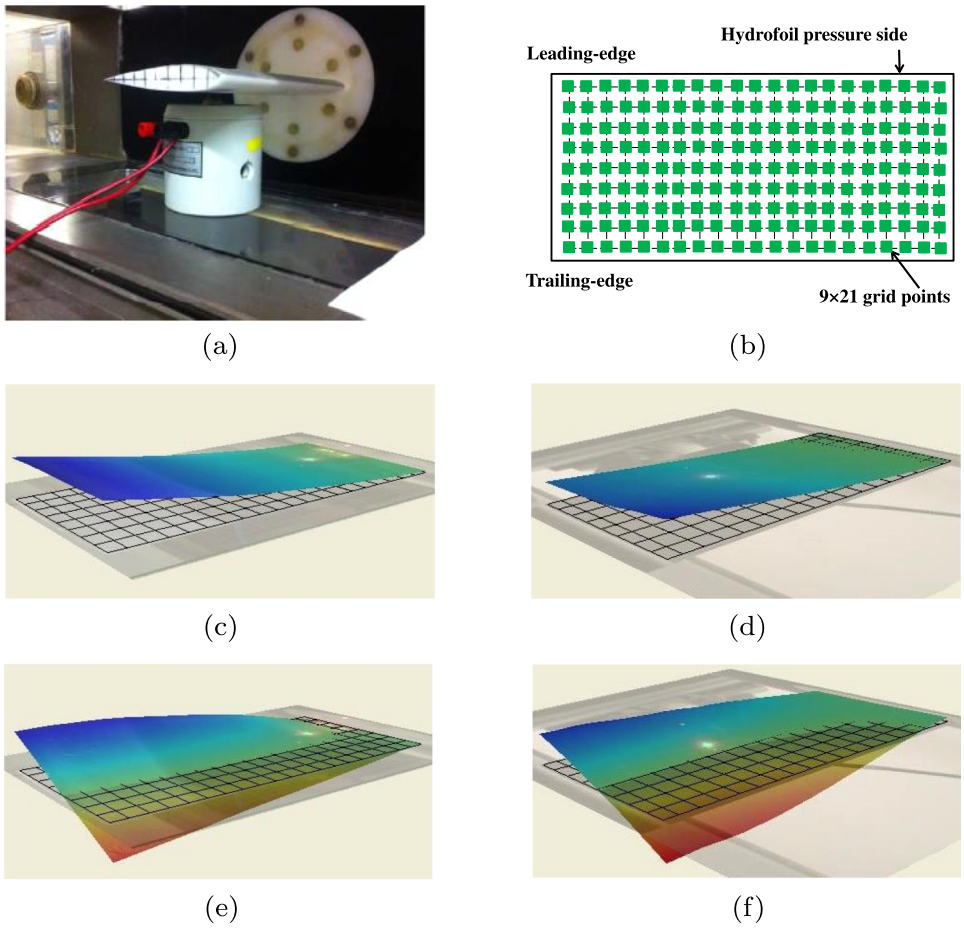


FIG. 2. Measured mode shapes obtained using (a) an electrodynamic shaker and the two LDV setups shown in Fig. 1(c) from IRENav, (b) scanning points (9×21 grid points) of the mesh on the foil surface; (c) 1st mode in-air, (d) 1st mode in-still-water, (e) 2nd mode in-air, and (f) 2nd mode in-still-water of the cantilevered NACA0015 POM hydrofoil.

assembly and disassembly of the mechanical mounting system supporting the hydrofoil. Based on repeated measurements under the same conditions, the uncertainty in the frequency measurement was approximately 2%.

For the measured frequency spectra of the foil in flowing water conditions, only two scanning points were used; the scanning points were located at a section that is 95% span length away from the clamped root, and at 0.1c or 0.8c from the foil leading-edge, and the reference laser was located at mid-chord, as shown in Fig. 1(d). To get time resolved data for the flow-induced vibration measurements, the velocimetry scanning head was held fixed to the ground. The hydrofoil vibrations were measured on the foil pressure side through the transparent bottom wall of the tunnel test section. The vibrations are measured between 0 and 2 kHz, and the sampling frequency and acquisition time of the vibrations are calculated by the LDV to obtain 3200 lines in this range. The vibration velocities were captured with sensitivities ranging from 10 to 1000 (mm/s)/V. The system provided the Fourier transform of the vibration velocity signal for each measurement point. The resultant velocity spectra were averaged using a user-defined number of acquisitions; specifically, the results shown in this work were computed by averaging 512 acquisitions per each measurement point. The frequency resolution for vibration measurements was $\Delta f = 0.625$ Hz. Experimental measurements of the natural flow-induced vibrations of the POM hydrofoil were collected for a range of reduced velocities ($\overline{U} = U/\omega_\theta b = 0.02$–$0.05$, i.e., $\mathrm{Re} = Uc/\nu_f = 3 \times 10^5$–$6 \times 10^5$, where $\omega_\theta = 2\pi f_\theta$ is the first in-air, or in-vacuum, natural twisting frequency, ν_f is the fluid kinematic viscosity), and α_o is the angle of attack ($\alpha_o = 0°$–$8°$).

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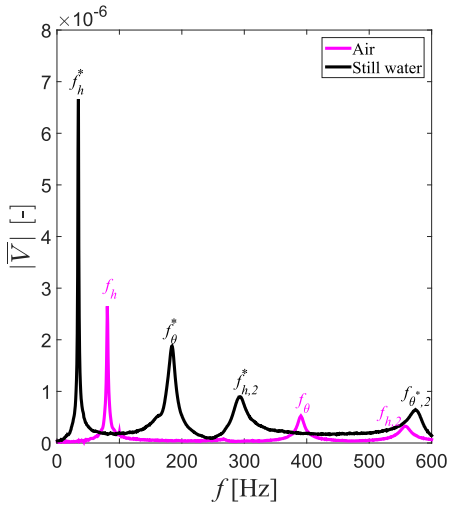


FIG. 3. Measured frequency responses of the foil's bending and twisting vibration velocities ($\overline{V}$) for the NACA0015 POM hydrofoil subject to known excitations via the electrodynamic shaker in-air and in-still-water conditions, as shown in Fig. 2(a). Note that f_h, f_θ, and $f_{h,2}$ are the first in-air natural bending frequency, the first in-air natural twisting frequency, and the second in-air natural bending frequency, respectively; f_h^*, f_θ^*, and $f_{h,2}^*$ are the corresponding in-water natural frequencies.

Figure 4 shows sample plots of the variation of the measured frequency spectra of the foil's bending and twisting velocity fluctuations with varying reduced velocities at $\alpha_o = 8°$ (Fig. 4(a)) and with varying angles of attack at $\overline{U} = 0.05$ (Fig. 4(b)). In general, the amplitude of the frequency spectra increased with increasing inflow velocity. The results in Fig. 4(a) show that the first in-water natural bending frequency was relatively insensitive to variations in the reduced velocity, but the first in-water natural twisting frequency increased with the reduced velocity. Figure 4(b) indicates that the in-water natural frequencies varied slightly with the angle of attack (α_o), but the variation was non-monotonic and was smaller than the variations observed in Fig. 4(a) for varying $\overline{U}$.

III. NUMERICAL MODEL

The numerical model shown in Fig. 5(a) was developed based on the experimental observation that the chordwise deformations of the current NACA0015 POM hydrofoil was very negligible compared to the spanwise bending, $\tilde{h}(z,t)$, and twisting, $\tilde{\theta}(z,t)$, deformations, where z is the spanwise coordinate with the origin at the root of the cantilevered hydrofoil, and t is time. To account for the spanwise bending and twisting flexibilities, the three-dimensional (3D) equations of motion (EOM) of the rectangular, cantilevered wing with an uniform cross section were expressed in generalized coordinates following the approach given in the work of Fung.[32] The generalized formulations are obtained by decomposing $\tilde{h}(z,t)$ and $\tilde{\theta}(z,t)$ via separation of variables in time and space,

$$\tilde{h}(z,t) = h(t)f(z), \tag{1}$$

$$\tilde{\theta}(z,t) = \theta(t)g(z), \tag{2}$$

TABLE I. Measured modal frequencies of the cantilevered NACA0015 POM hydrofoil in-air and in-still-water via impulse excitations using an electrodynamic shaker, as shown in Fig. 2(a).

Modal frequency (Hz)	Air	Still water
1st mode	81	34
2nd mode	390	184
3rd mode	557	292

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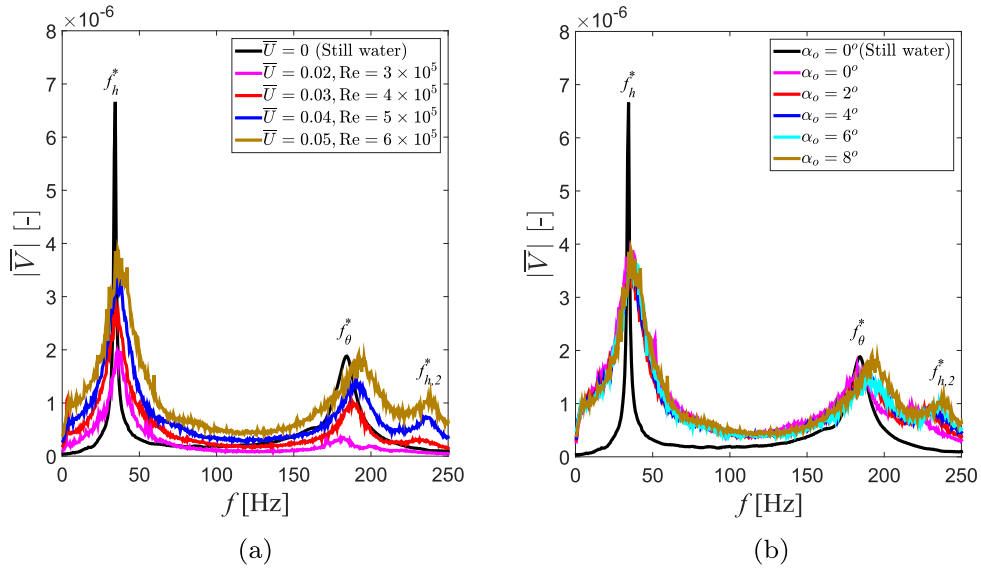


FIG. 4. The variation of the measured frequency spectra of the foil's bending and twisting velocity fluctuations for the NACA0015 POM hydrofoil in water with (a) varying reduced velocity at $\alpha_o = 8°$ and (b) varying angle of attack at $\overline{U} = 0.05$. The in-still-water ($\overline{U} = 0$) result is obtained using an electrodynamic shaker shown in Fig. 2(a), while the in-flowing-water ($\overline{U} > 0$) results are obtained by natural flow-induced vibrations.

where $h(t)$ and $\theta(t)$, as shown in Fig. 5(a), represent the time varying bending and twisting deformations at the free tip of the foil defined about the elastic axis (EA). $f(z)$ and $g(z)$, as shown in Fig. 5(b), represent non-dimensional spanwise shape functions for the first bending and first twisting modes, respectively.

The shape functions shown in Fig. 5(b) were obtained by taking the mean value of the experimentally measured mode shapes (shown in Fig. 2). Additionally, the theoretical shape functions of a cantilevered beam were calculated by using Eqs. (3) and (4) (from Fung[32]) were compared with the measured functions in Fig. 5(b). The good agreement between the measured and theoretical shape functions supports the assumption that the foil behaves like a theoretical rectangular beam/plate that is rigid in the chordwise direction,

$$f(\overline{z}) = \frac{\cosh(1.875\overline{z}) - \cos(1.875\overline{z}) - 0.734\left[\sinh(1.875\overline{z}) - \sin(1.875\overline{z})\right]}{2}, \tag{3}$$

$$g(\overline{z}) = \sin\left(\frac{\pi}{2}\overline{z}\right), \tag{4}$$

where $\overline{z} = z/s$ is the non-dimensional span length.

In Fig. 5, the inflow velocity, which is aligned to the X-axis, is U. The geometric angle of attack is defined as α_o, the effective angle of attack is defined as $\alpha_{\mathrm{eff}} = \alpha_o - \theta$, and θ is defined positive nose down about the EA. $\tilde{K}_{s,h}(=\tilde{m}\omega_h^2)$ and $\tilde{K}_{s,\theta}(=\tilde{I}_\theta\omega_\theta^2)$ represent the spanwise bending and twisting stiffnesses, respectively. $\tilde{C}_{s,h}(=2\tilde{m}\omega_h\zeta_{s,h})$ and $\tilde{C}_{s,\theta}(=2\tilde{I}_\theta\omega_\theta\zeta_{s,\theta})$ are, respectively, the spanwise bending and twisting damping values. $\tilde{m}$ is the generalized solid mass, $\tilde{I}_\theta = \tilde{m}r_\theta^2 b^2$ is the generalized solid mass moment of inertia about EA (EA is located at a distance ab in the downstream direction from the mid-chord), and r_θ is the non-dimensional radius of gyration. $\omega_h = 2\pi f_h$ and $\omega_\theta = 2\pi f_\theta$ are, respectively, the first in-air (in-vacuum) natural bending and twisting frequencies, which are taken to be the same as the in-air natural frequencies because of the low density of air. $\zeta_{s,h}$ and $\zeta_{s,\theta}$ are, respectively, the structural bending and twisting damping coefficients. The detailed derivation of generalized formations (which are formulated by integrating the sectional solid and fluid forces along the span to obtain the generalized mass, damping, and stiffness matrices, as well as the hydrodynamic forcing vector) can be found in Akcabay *et al.*[33]

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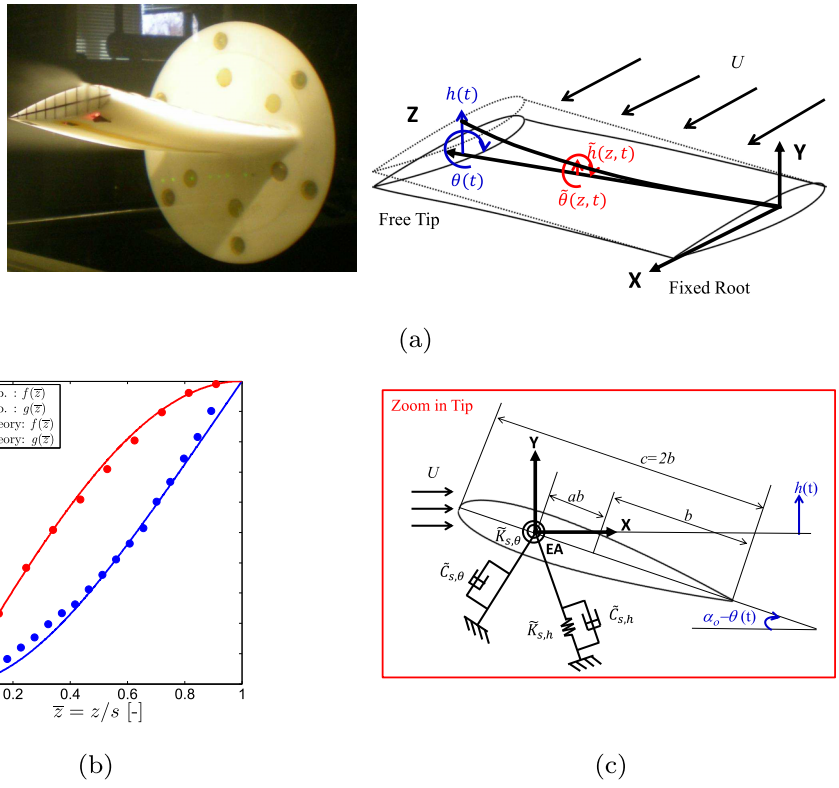


FIG. 5. (a) A sample photograph of the flexible POM hydrofoil undergoing hydroelastic deformations and definition of a cantilevered, flexible, rectangular hydrofoil with spanwise bending and twisting flexibilities, (b) measured and theoretically estimated normalized spanwise bending, $f(z)$, and twisting, $g(z)$, shape functions, and (c) a two-degree-of-freedom (2-DOF) structural model representing of the free tip response of the cantilever, flexible hydrofoil.

A. Non-dimensional governing equations

The viscous fluid model solved the incompressible, unsteady Reynolds-averaged Navier-Stokes (URANS) equations (Eqs. (5) and (6)) with the $k-\omega$ shear stress transport ($k-\omega$ SST) turbulence model. Note that the governing equations can be consistently non-dimensionalized using the characteristic dimensional parameters defined in Table II, where ρ_f is the fluid density,

$$\frac{\partial \overline{u}_i}{\partial \overline{x}_i} = 0, \tag{5}$$

$$\frac{\partial \overline{u}_i}{\partial \overline{t}} + \overline{u}_j \frac{\partial \overline{u}_i}{\partial \overline{x}_j} = -\frac{\partial \overline{p}}{\partial \overline{x}_i} + \frac{\partial}{\partial \overline{x}_j}\left[\left(\frac{2\overline{U}}{\mathrm{Re}} + \overline{\nu}_t\right)\left(\frac{\partial \overline{u}_i}{\partial \overline{x}_j} + \frac{\partial \overline{u}_j}{\partial \overline{x}_i}\right) + \frac{2}{3}\overline{K}\delta_{ij}\right], \tag{6}$$

TABLE II. Characteristic dimensional parameters.

Variable	Unit	Characteristic parameter
Length	(m)	b
Mass	(kg)	$\pi\rho_f b^2 s$
Time	(s)	$1/\omega_\theta$
Velocity	(m/s)	$\omega_\theta b$
Lift	(N)	$\pi\rho_f b^3\, s\omega_\theta^2$
Moment	(Nm)	$\pi\rho_f b^4\, s\omega_\theta^2$
Pressure	(Pa)	$\rho_f \omega_\theta^2 b^2$

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where $\overline{u}$ is the non-dimensional local fluid velocity; $\overline{x}$ is the non-dimensional position; $\overline{t}$ is the non-dimensional time; $\overline{p}$ is the non-dimensional total pressure; $\mathrm{Re} = Uc/\nu_f = 2\overline{U}(\omega_\theta b^2)/\nu_f$ is the Reynolds number, where ν_f is the fluid kinematic viscosity; $\overline{\nu_t} = \nu_t/(\omega_\theta b^2)$, where ν_t is the turbulent (eddy) kinematic viscosity shown in Eq. (7); $\overline{K} = K/(\omega_\theta^2 b^2)$, where, K is the turbulence kinetic energy from Eq. (8); and $\delta_{ij} = 1$ for $i = j$ or $\delta_{ij} = 0$ for $i \neq j$,

$$\nu_t = \frac{a_1 K}{\max(a_1, \mathrm{W}, \mathrm{SF}_2)}, \tag{7}$$

where $a_1 = 0.31$, W is the specific dissipation rate from Eq. (9), and F_2 is second blending function, which restricts the limiter to the wall boundary layer, as the underlying assumptions are not correct for free shear flows. S is an invariant measure of the strain rate,

$$\frac{\partial K}{\partial t} + u_j \frac{\partial K}{\partial x_j} = P_K - \beta^* K W + \frac{\partial}{\partial x_j}\left[(\nu_f + \sigma_K \nu_t)\frac{\partial K}{\partial x_j}\right], \tag{8}$$

$$\frac{\partial W}{\partial t} + u_j \frac{\partial W}{\partial x_j} = \alpha S^2 - \beta W + \frac{\partial}{\partial x_j}\left[(\nu_f + \sigma_W \nu_t)\frac{\partial W}{\partial x_j}\right] + 2(1 - F_1)\sigma_{W2}\frac{1}{W}\frac{\partial K}{\partial x_j}\frac{\partial W}{\partial x_j}, \tag{9}$$

where P_K is the production term, F_1 is a blending function, β^*, β, α, σ_K, σ_W, and σ_{W2} are constant values, as defined in Menter.[34]

The viscous fluid-structure interaction (FSI) response of the hydrofoil was modeled by using the loose hybrid coupled (LHC) method to couple an URANS fluid solver and a two-degree-of-freedom (2-DOF) generalized equation of motion model (EOM) for the model shown in Fig. 5. The viscous coupled generalized EOM in Eq. (10) was solved using a semi-implicit Crank-Nicolson method,

$$\tilde{\mathbf{M}}_s \ddot{\tilde{\mathbf{X}}}_{n+1} + \tilde{\mathbf{C}}_s \dot{\tilde{\mathbf{X}}}_{n+1} + \tilde{\mathbf{K}}_s \tilde{\mathbf{X}}_{n+1} - (\tilde{\mathbf{F}}_{\mathrm{FSI}})_{n+1} = (\tilde{\mathbf{F}}_{\mathrm{CFD}})_n - (\tilde{\mathbf{F}}_{\mathrm{FSI}})_n. \tag{10}$$

In Eq. (10), the subscript n is the time step index. The left hand side of Eq. (10) is discretized implicitly and the right hand side of Eq. (10) is discretized explicitly. $\tilde{\mathbf{X}} = [\tilde{h}/b, \tilde{\theta}]^\top$, $\dot{\tilde{\mathbf{X}}}$, and $\ddot{\tilde{\mathbf{X}}}$ are the non-dimensional generalized displacement, velocity, and acceleration vectors. $\tilde{\mathbf{M}}_s$, $\tilde{\mathbf{C}}_s$, and $\tilde{\mathbf{K}}_s$ (Eqs. (11)-(13)) are, respectively, the non-dimensional generalized solid inertial, damping, and stiffness matrices,

$$\tilde{\mathbf{M}}_s = \mu \begin{bmatrix} S_{ff} & x_\theta S_{fg} \\ x_\theta S_{fg} & r_\theta^2 S_{gg} \end{bmatrix}, \tag{11}$$

$$\tilde{\mathbf{C}}_s = \mu \begin{bmatrix} 2\Omega\zeta_{s,h} S_{ff} & 0 \\ 0 & 2r_\theta^2 \zeta_{s,\theta} S_{gg} \end{bmatrix}, \tag{12}$$

$$\tilde{\mathbf{K}}_s = \mu \begin{bmatrix} \Omega^2 S_{ff} & 0 \\ 0 & r_\theta^2 S_{gg} \end{bmatrix}, \tag{13}$$

where $\mu = \tilde{m}/(\pi \rho_f b^2 s)$ is the solid-to-fluid added mass ratio. x_θ is the non-dimensional distance from the EA to the center of gravity (CG). S_{ff}, S_{fg}, and S_{gg} are the integrated shape functions,

$$S_{ff} = \int_0^1 f^2(\overline{z}) d\overline{z}, \tag{14}$$

$$S_{fg} = \int_0^1 f(\overline{z}) g(\overline{z}) d\overline{z}, \tag{15}$$

$$S_{gg} = \int_0^1 g^2(\overline{z}) d\overline{z}. \tag{16}$$

$\tilde{\mathbf{F}}_{\mathrm{CFD}} = [\tilde{L}_{\mathrm{CFD}}, \tilde{M}_{\mathrm{CFD}}]^\top = [L_{\mathrm{CFD}} S_{ff}, M_{\mathrm{CFD}} S_{gg}]^\top$ in the right hand side of Eq. (10) is the non-dimensional generalized viscous fluid force vector, which is obtained by solving the URANS equations using the commercial CFD software of ANSYS-CFX,[35] based on the instantaneous foil geometry obtained by solving Eq. (10) at the previous time step.

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The LHC method delivers numerically stable solutions by subtracting the potential flow estimation of the generalized FSI force vector, $\tilde{\mathbf{F}}_{\mathrm{FSI}}$, as derived by Theodorsen[2] (Eq. (17)), from both the left and right hand sides of the EOM (Eq. (10)),

$$\tilde{\mathbf{F}}_{\mathrm{FSI}} = -\tilde{\mathbf{M}}_f\ddot{\tilde{\mathbf{X}}} - \tilde{\mathbf{C}}_f\dot{\tilde{\mathbf{X}}} - \tilde{\mathbf{K}}_f\tilde{\mathbf{X}}. \tag{17}$$

In Eq. (17), $\tilde{\mathbf{M}}_f$, $\tilde{\mathbf{C}}_f$, and $\tilde{\mathbf{K}}_f$ are, respectively, the non-dimensional generalized Theodorsen's potential flow approximation of the fluid inertial, damping, and stiffness matrices,[2]

$$\mathbf{M}_f = \begin{bmatrix} S_{ff} & -aS_{fg} \\ -aS_{fg} & \left(\frac{1}{8} + a^2\right)S_{gg} \end{bmatrix}, \tag{18}$$

$$\mathbf{C}_f = \overline{U}\begin{bmatrix} 2C(k)S_{ff} & (1 - 2C(k)d)S_{fg} \\ -2C(k)eS_{fg} & d[2C(k)e - 1]S_{gg} \end{bmatrix}, \tag{19}$$

$$\mathbf{K}_f = \overline{U}^2\begin{bmatrix} 0 & 2C(k)S_{fg} \\ 0 & -2C(k)eS_{gg} \end{bmatrix}, \tag{20}$$

where $C(k)$ is the Theodorsen's circulation function, which accounts for the circulatory load on the foil induced by the shed vortices in the wake; $k = \omega b/U$ is the reduced foil oscillation frequency. To get the initial k, the inviscid coupled EOM (Eq. (10)) is iterated first, and the viscous coupled EOM uses the same k from the solution of the inviscid coupled EOM.

In linearized, potential flow models such as the approach of Theodorsen[2] and Weissinger,[36] the flow is assumed to be mostly attached and the net lift is assumed to act at the aerodynamic center (AC), i.e., the center of pressure (CP) is assumed to coincide with the AC. Hence, e is assumed to take the value of $a + \frac{1}{2}$ (the AC is at quarter-chord from leading-edge of a symmetric thin foil). They also assume the trailing-edge vortex shedding frequency to be the same as the foil oscillation frequency and the vortices to convect downstream without any dissipation. These inviscid derivations also assume the induced downwash, which is caused by the shed vortices, to act at three quarter-chord ($3c/4$) from the foil's leading-edge, which is located at distance db in the downstream direction from the EA, i.e., $d = a - \frac{1}{2}$. However, CP, where the lift actually acts, changes with operating conditions, particularly for thick and rounded nose sections common to many maritime applications. In addition, the vortex shedding frequency may differ from the foil oscillation frequency and may contain multiple frequencies. Depending on the angle of attack and inflow velocity, the wake may have a certain thickness normal to the inflow direction. The vortices may start to shed from the foil's leading-edge if the flow is fully separated and may interact with foil motions; viscous dissipation may occur; and the effective induced velocity on the foil by the shed vortices may not necessarily act at $3c/4$ from the foil's leading-edge. Hence, $\tilde{\mathbf{F}}_{\mathrm{CFD}}$ is needed on the right hand side of Eq. (10) to account for viscous effects.

The characteristic non-dimensional parameters governing the FSI response of a flexible hydrofoil are listed in Table III, where $\tilde{S}_\theta = \tilde{m}x_\theta b$ is the generalized static imbalance, and $\omega = 2\pi f$ is the foil's free oscillation frequency. It should be noted that for lightweight foils operating in dense fluids such that $\sqrt{\mu} < 1$, the relative contribution of the fluid inertial and damping terms will be higher than that of the corresponding solid terms.

B. Prediction of the total loss factor

The total loss factor is equal to two times the total damping coefficient (i.e., $\eta_T = 2\zeta_T$), which is the sum of the solid loss factor (η_s) and the fluid loss factor (η_f),

$$\eta_T = \eta_s + \eta_f. \tag{21}$$

The theoretical inviscid, *uncoupled mode* (UM), fluid loss factors are given in Eqs. (22) and (23), which is based on putting zeros to the off-diagonal terms in $\mathbf{M}_f$, $\mathbf{C}_f$, and $\mathbf{K}_f$ in Eqs. (18)-(20), and Blake and Maga[6] used these factors in their analysis.

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TABLE III. Characteristic non-dimensional parameters of a cantilevered flexible hydrofoil in viscous fluid flow with spanwise bending and twisting flexibility.

Parameter	Symbol	Definition	Physical meaning
Reynolds number	Re	$2Ub/\nu_f$	Ratio between fluid inertial force and fluid viscous force
Solid-to-fluid added mass ratio	μ	$\tilde{m}/(\pi\rho_f b^2 s)$	Ratio between solid inertial force and fluid inertial force for bending motion
Reduced velocity	$\overline{U}$	$U/(\omega_\theta b)$	Ratio between fluid convection frequency and solid first in-air natural twisting frequency, also the ratio between the fluid disturbing force to the solid restoring force
Solid bending to twisting frequency ratio	Ω	ω_h/ω_θ	Ratio between first in-vacuum (or in-air) natural bending and twisting frequency
Solid damping coeff.	$\zeta_{s,h}$, $\zeta_{s,\theta}$	$\tilde{C}_{s,h}/(2\tilde{m}\omega_h)$, $\tilde{C}_{s,\theta}/(2\tilde{I}_\theta\omega_\theta)$	Solid bending and twisting damping coefficients
Elastic axis location	a	...	Non-dimensional distance from mid-chord to foil elastic axis (EA), positive if EA is aft of the mid-chord
Center of gravity location	x_θ	$\tilde{S}_\theta/(\tilde{m}b)$	Non-dimensional distance from the EA to the foil center of gravity (CG), positive for CG aft of the EA
Radius of gyration	r_θ	$\sqrt{\tilde{I}_\theta/(\tilde{m}b^2)}$	Non-dimensional radius of gyration about the EA
Reduced frequency	k	$\omega b/U$	Ratio between solid oscillation frequency and fluid convection frequency

$$\eta_{\mathrm{f,h-UM}} = \frac{2C(k)\overline{U}}{\Omega\sqrt{\mu(\mu+1)}}, \tag{22}$$

$$\eta_{\mathrm{f},\theta-\mathrm{UM}} = \frac{d[2C(k)e-1]\overline{U}}{\sqrt{\left[\mu r_\theta^2 - 2C(k)e\overline{U}^2\right]\left[\mu r_\theta^2 + \left(\frac{1}{8}+a^2\right)\right]}}. \tag{23}$$

The mobility peak power down method presented in Cremer, Heckl, and Petersson,[37] as shown in Eq. (24), was used to calculate the total loss factor ($\eta_T = 2\zeta_T$), by processing the frequency spectra of both the experimental measurements and viscous FSI simulations. Note that the experimental results by IRENav are only in terms of frequency spectra; therefore the classical logarithmic decrement method cannot be used to deduce the damping factor. The mobility peak power down method to calculate the total loss factor is shown in Eq. (24) with the definitions given in Fig. 6 and

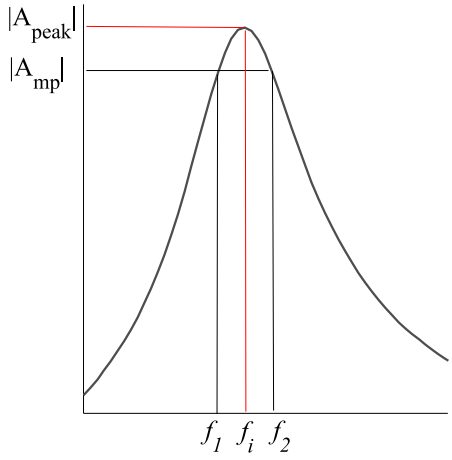


FIG. 6. Illustration of the mobility peak power down method used to calculate the total loss factors based on the frequency spectrum.

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detailed below in the text,

$$\eta_T = \frac{f_2 - f_1}{f_i\sqrt{p-1}}, \tag{24}$$

where p is the number of the reduction of the mobility such that $|A_{mp}| = |A_{peak}|/\sqrt{p}$. In Eq. (24) and Fig. 6, f_i is the frequency of interest (e.g., bending and twisting natural frequencies, or the vortex shedding frequency) with peak amplitude (i.e., $|A_{peak}|$). f_1 and f_2 correspond to the frequencies on the left and right, respectively, at the amplitude $|A_{mp}|$ (i.e., pth mobility peak amplitude, on each side of f_i), as shown in Fig. 6. In this study, $p = 5$, following recommendations in Reese,[7] was used when processing both experimental and viscous FSI results.

C. Numerical model setup

Numerical simulations were performed on cantilevered, NACA0015 hydrofoils made of three different materials: polyacetate (i.e., POM), aluminum (i.e., Al), and stainless steel (i.e., steel).

Table IV lists a comparison of the key parameters for the current NACA0015 hydrofoils with the stainless steel and aluminum hydrofoils tested by Blake and Maga[6] and by Reese,[7] respectively. The in-air natural bending and twisting frequencies (f_h and f_θ) of the current NACA0015 Al and steel hydrofoils in Table IV were calculated using Eqs. (25) and (26), following the theoretical approach given in Leissa,[38] which was derived for rectangular, homogeneous, isotropic, cantilevered plates,

$$\omega_h = 2\pi f_h = \frac{\beta_1}{s}\sqrt{\frac{E_s(\tau/s)^2}{12(1-\nu_s^2)\rho_s}}, \tag{25}$$

$$\omega_\theta = 2\pi f_\theta = \frac{\beta_2}{s}\sqrt{\frac{E_s(\tau/s)^2}{12(1-\nu_s^2)\rho_s}}, \tag{26}$$

where τ is the maximum thickness of the foil. $\beta_1 = 2.88$ and $\beta_2 = 13.84$ were determined by matching the in-air natural frequencies of the NACA0015 POM hydrofoil measured at the French Naval Academy Research Institute (IRENav). The assumed solid damping coefficients ($\zeta_{s,h}$, $\zeta_{s,\theta}$)

TABLE IV. Key physical and geometrical parameters of the current NACA0015 hydrofoils tested in this study, as well as the stainless steel (i.e., steel) and aluminum (i.e., Al) hydrofoils tested by Blake and Maga[6] and Reese,[7] respectively.

Model	Blake and Maga, expt.	Reese, expt.	Current expt. and num.
Foil shape	Bullet	NACA66	NACA0015
Material	Steel	Al	POM, Al, Steel
ρ_s (kg/m^3)	N/A	N/A	1420, 2700, 8010
E_s (Gpa)	N/A	N/A	2.9, 69, 203
ν_s (-)	N/A	N/A	0.35, 0.33, 0.3
$\sqrt{\mu}$ (-)	0.82	0.47	0.43, 0.59, 1.02
f_h (Hz)	114	319	81, 284, 274
f_θ (Hz)	897	1549	390, 1374, 1324
$\zeta_{s,h}$ (-)	N/A	N/A	0.02, 0.002, 0.001
$\zeta_{s,\theta}$ (-)	N/A	N/A	0.02, 0.002, 0.001
a (-)	0	0	−0.28
x_θ (-)	N/A	N/A	0.13
r_θ (-)	N/A	N/A	0.43
s (m)	0.508	0.1	0.191
c (m)	0.07	0.05	0.1
α_o (°)	0	−4, 0, 4, 10	2, 8, 15, 17, 20
Re (-)	1×10^5–5×10^5	5.6×10^4–5.6×10^5	4×10^5–2.1×10^6
$\overline{U}$ (-)	0.007–0.04	0.004–0.04	0.03–0.1

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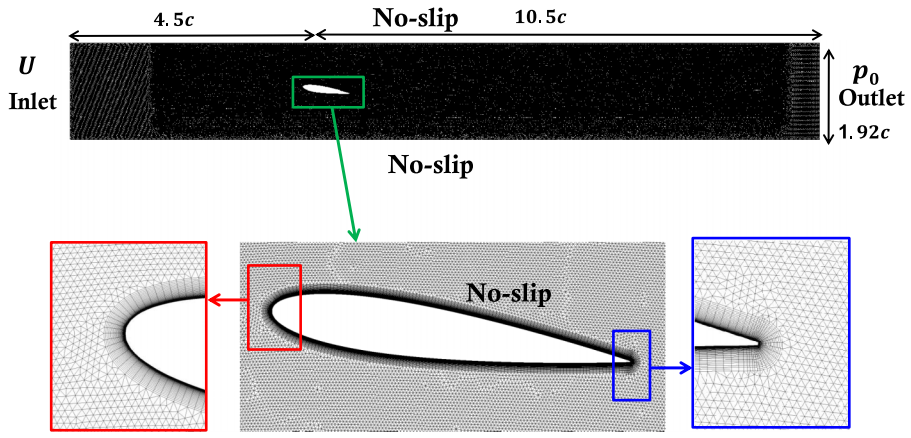


FIG. 7. NACA0015 CFD (medium) mesh with 6.0×10^5 nodes and 5.9×10^5 elements and prescribed boundary conditions.

for NACA0015 POM, Al, and steel hydrofoils (0.02, 0.002, and 0.001, respectively) given in Table IV are typical values for these materials and account for the higher material damping typically observed for more flexible materials. The solid damping coefficient was assumed to be constant for each material, so any variation in the total loss factor with flow conditions is regarded to result from changes in the hydrodynamic damping only. It should be noted that the small difference in solid damping is almost inconsequential, as the fluid damping is about an order of magnitude higher for the marine application cases considered in this paper, as will be shown in Sections IV–VI, and hence dominates the vibration response in water.

The numerical mesh shown in Fig. 7 was used to solve the fluid dynamic equations. In Fig. 7, an unstructured mesh was used everywhere, apart from the hydrofoil surface; a structured boundary layer mesh was used near the foil surface, which satisfied $y^+ \approx 1$; and the mesh was refined near the foil's leading-edge, trailing-edge, and the wake region to capture the flow details. The boundary conditions of the CFD mesh domain were

- uniform inlet boundary condition with 2% turbulence intensity at the left edge of the domain,
- prescribed pressure boundary condition at the right edge of the domain,
- no-slip and no-penetration boundary conditions at the foil interface, top edge, and bottom edge of the domain, which matches the experimental water tunnel dimensions to account for the flow confinement effects.

At each time step, the mesh elements were deformed to conform to the hydrofoil geometry according to the foil motions obtained by using the viscous loose hybrid coupled (LHC) method described in Section III A. (Readers should refer to ANSYS-CFX[35] for details about the mesh deformation algorithm.)

The numerical simulations of the flexible hydrofoils were initialized with zero initial twisting and bending deformations from the steady-state solution of flow around a stationary hydrofoil; the flow was assumed to be fully turbulent, where the URANS model is valid; and the hydrofoil was assumed to be deeply submerged so free surface and cavitation effects are not considered.

D. Convergence study

The convergence studies of numerical mesh and time step sizes were conducted on the cantilevered flexible NACA0015 POM hydrofoil ($\sqrt{\mu} = 0.43$) at $\alpha_o = 8°$ and $\overline{U} = 0.05$ (Re $= 6\times10^5$).

For a mesh convergence study, Fig. 8 compares the predicted time-histories of the natural flow-induced spanwise bending and twisting deformations at the tip of a cantilevered flexible NACA0015 POM hydrofoil while using $\Delta t = T_h^*/360$ with $y^+ = 1$ for three different mesh sizes: "coarse mesh" $= 3.5\times10^5$ nodes and 3.4×10^5 elements, "medium mesh" $= 6.0\times10^5$ nodes and 5.9×10^5 elements, and "fine mesh" $= 11.0\times10^5$ nodes and 10.8×10^5 elements. $T_h^* = 1/f_h^*$ is the

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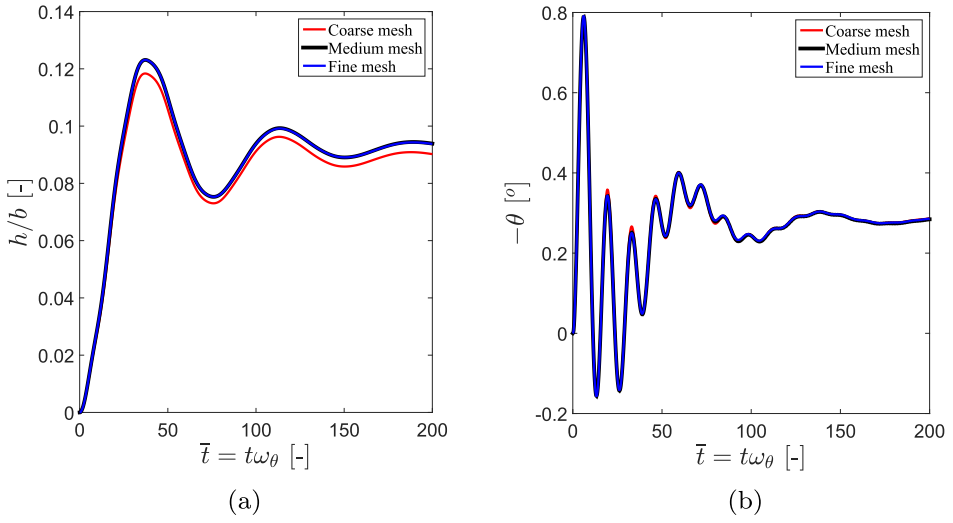


FIG. 8. Mesh convergence study of the natural flow-induced spanwise (a) bending deformations and (b) twisting deformations at the tip of a cantilevered flexible NACA0015 POM hydrofoil in water while using $\Delta t = T_h^*/360$ with $y^+ = 1$, $\alpha_o = 8°$, $\sqrt{\mu} = 0.43$, and $\overline{U} = 0.05$ ($\mathrm{Re} = 6\times10^5$). The "coarse mesh" has 3.5×10^5 nodes and 3.4×10^5 elements, "medium mesh" has 6.0×10^5 nodes and 5.9×10^5 elements, and "fine mesh" has 11.0×10^5 nodes and 10.8×10^5 elements. $T_h^* = 1/f_h^*$ is the first in-water natural bending period of the flexible hydrofoil.

first in-water natural bending period of the flexible hydrofoil. Note that the superscript * is used to denote the in-water frequencies. The results in Fig. 8 show that the predicted time-histories of the bending and twisting deformations with the medium mesh were sufficiently close to the predictions evaluated with the fine mesh. Therefore, the medium mesh was used to calculate the viscous FSI responses for the results shown in the rest of this paper.

For a time step size convergence study, Fig. 9 compares the predicted time-histories of the natural flow-induced spanwise bending and twisting deformations at the tip of a cantilevered flexible NACA0015 POM hydrofoil while using three different time step sizes, $\Delta t = T_h^*/180$, $T_h^*/360$, and $T_h^*/720$ with the "medium mesh" at $y^+ = 1$, $\alpha_o = 8°$, $\sqrt{\mu} = 0.43$, and $\overline{U} = 0.05$ ($\mathrm{Re} = 6\times10^5$).

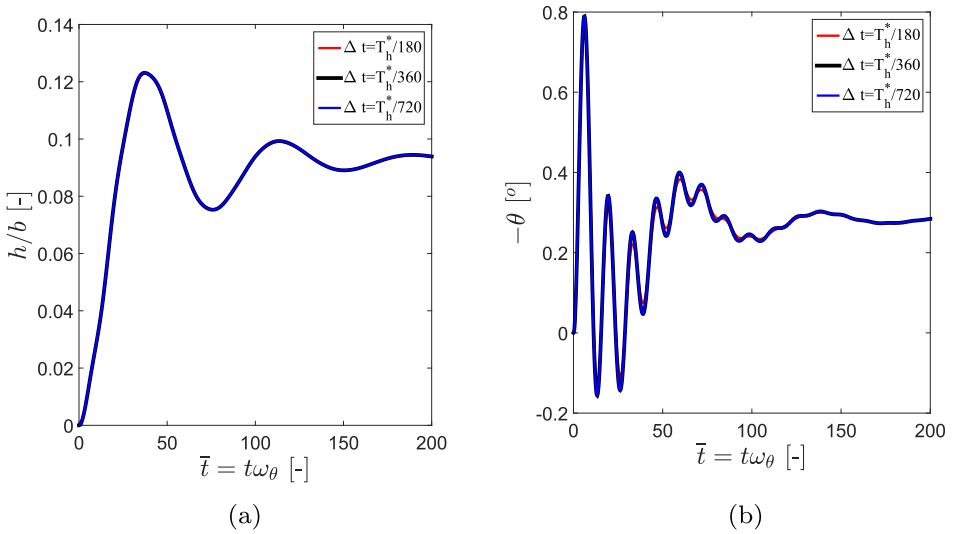


FIG. 9. Time step size convergence study of the natural flow-induced spanwise (a) bending deformations and (b) twisting deformations at the tip of a cantilevered flexible NACA0015 POM hydrofoil in water while using the "medium mesh" with $y^+ = 1$, $\alpha_o = 8°$, $\sqrt{\mu} = 0.43$, and $\overline{U} = 0.05$ ($\mathrm{Re} = 6\times10^5$). Note that $T_h^* = 1/f_h^*$ is the first in-water natural bending period of the flexible hydrofoil.

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Note in Fig. 9 that the predictions obtained using the two smallest time steps were consistent. Consequently, all the simulations shown next used a time step size of $\Delta t = T_h^*/360$. Comparisons of the numerical predictions and experimental measurements will be shown later in Sections III E, IV, and V.

E. Validation study

Validation studies of the viscous 2-DOF FSI predictions obtained using the LHC method with experimental measurements and other numerical/analytical predictions of rigid and flexible hydrofoils in both steady-state and dynamic conditions can be found in previous works by the authors. Good comparisons were observed between the predicted and measured (1) steady-state hydrodynamic load coefficients for both rigid and flexible hydrofoils, including in the post-stall regime,[39,40] (2) hydrodynamic load coefficients for a rigid hydrofoil subject to controlled (forced) pitch motion in fully wetted[41,42] and cavitating conditions,[41] (3) flutter and divergence velocities, and flutter frequencies for airfoils and hydrofoils across a range of solid-to-fluid-added-mass ratios,[43] and (4) cavitation patterns, maximum cavity length, and frequency spectra of the deformation responses for flexible hydrofoils in unsteady cavitating flows.[33,41,44]

1. Steady state validation study

As an additional check of the validity of the current experimental and numerical model results, the measured and predicted static lift coefficient (C_L), drag coefficient (C_D), and moment coefficient at the aerodynamic center ($C_{Mo,\mathrm{AC}}$) with varying angles of attack (α_o) for a rigid NACA0015 hydrofoil are plotted together with published experimental results from Jacobs and Sherman[45] in Fig. 10. For simplicity, the current experiments, as explained in Section II, are denoted with "Exp" in figure legends. The setup of the rigid NACA0015 hydrofoil in the current study is the same as given in Sections II and III C, but for this particular case, the foil was made of a stainless steel and the axis of rotation was at $0.25c$ from the foil's leading-edge.

Jacobs and Sherman used a NACA0015 airfoil with aspect ratio (AR) of 6, while the current experiments by the French Naval Academy Research Institute (IRENav) used a NACA0015 hydrofoil with AR = 3.82. Notice that AR = $2s/c$ for cantilevered configurations. Figure 10 also includes the predicted static lift coefficients from the inviscid thin-airfoil theory given in Eq. (27).

$$C_L = 2\pi\alpha_o. \tag{27}$$

The theoretical steady-state drag coefficient given in Fig. 10(b) is calculated using Eq. (28), where C_{D_o} is the skin friction drag coefficient based on 1/7 power law, and C_{D_i} is the induced drag

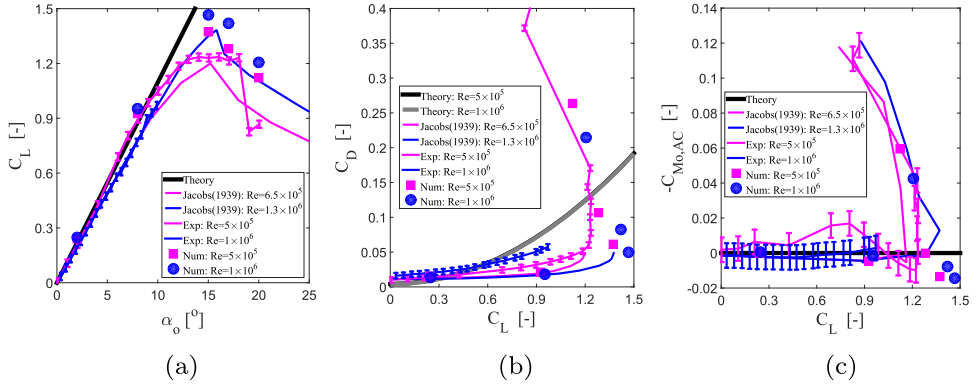


FIG. 10. (a) C_L vs. α_o, (b) C_L vs. C_D, and (c) $C_{Mo,\mathrm{AC}}$ vs. C_L plots for a rigid NACA0015 foil at steady-state. The aspect ratio (AR = $2s/c$ for cantilevered configurations) was 6 for the experiments by Jacobs and Sherman[45] and was 3.82 for the current experiments by the French Naval Academy Research Institute (IRENav). The error bars of experimental measurements indicate the uncertainties of the measurements for $C_L = \pm 0.02$, $C_D = \pm 0.004$, and $C_{Mo,\mathrm{AC}} = \pm 0.007$.

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coefficient for a foil with an elliptic lift coefficient distribution.

$$C_D = C_{D_o} + C_{D_i} = 0.0576\mathrm{Re}^{-1/5} + C_L^2/(\pi \mathrm{AR}). \tag{28}$$

The theoretical moment coefficients at the aerodynamic center, $C_{Mo,\mathrm{AC}}$, given in Fig. 10(c) are zero because the inviscid theory assumes the center of pressure (CP) to be located at the AC.

As shown in Fig. 10, the current experimental and numerical results are similar to experimental results of Jacobs and Sherman[45] for similar Reynolds numbers (Re). The results in Fig. 10 show that the measured and predicted lift coefficients are less than those of inviscid theory. The lift coefficient decreased from the maximum, when the angle of attack exceeded the stall angle of about $\alpha_o = 16°$–$18°$. It is noteworthy that the moment coefficient gets increasingly more different from the inviscid theory with the development of a trailing-edge vortex (TEV) with higher angles of attack (shown later in Fig. 20), and the corresponding shift of the CP position towards the mid-chord.

Figure 10 shows a good agreement between the predicted and measured hydrodynamic load coefficients for $\alpha_o \leq 15°$, the pre-stall regime, which suggest that the fully turbulent flow assumption is valid for the range Re of considered. Note that some differences are observed between the predicted and measured stall point, and the post-stall behavior, where URANS may not be sufficient because the flow becomes dominated by large-scale transient vortices, and 2D flow assumption breaks down. Based on previous experimental measurements from Ducoin, Astolfi, and Sigrist[29] and Ducoin and Young[40], the flow is fully turbulent for $\mathrm{Re} \geq 4 \times 10^5$, and transitional for $\alpha_o \leq 6°$ and $\mathrm{Re} < 4 \times 10^5$. In the current numerical simulations, the flow was assumed to be fully turbulent, as transition will significantly increase the complication of the modeling, but will affect only a very narrow range in the collected data space. For proper prediction of a laminar separation bubble (LSB) and transitional flow dynamics for stall and post-stall responses, a large-eddy simulation (LES), detached eddy simulation (DES), or even direct numerical simulation (DNS)[46] model coupled with suitable three-dimensional (3D) structural model is needed, which is outside the scope of the current work.

2. *Unsteady validation study*

For the viscous FSI simulations of the unsteady flow, Fig. 11 compares the variation of the predicted and measured frequency spectra via the fast Fourier transform (FFT) method with the moving average function of the natural flow-induced tip bending and twisting velocity fluctuations ($\overline{V}$) with varying reduced velocity ($\overline{U}$ = 0.03 and 0.05, Re = 4×10^5 and 6×10^5) for a cantilevered flexible

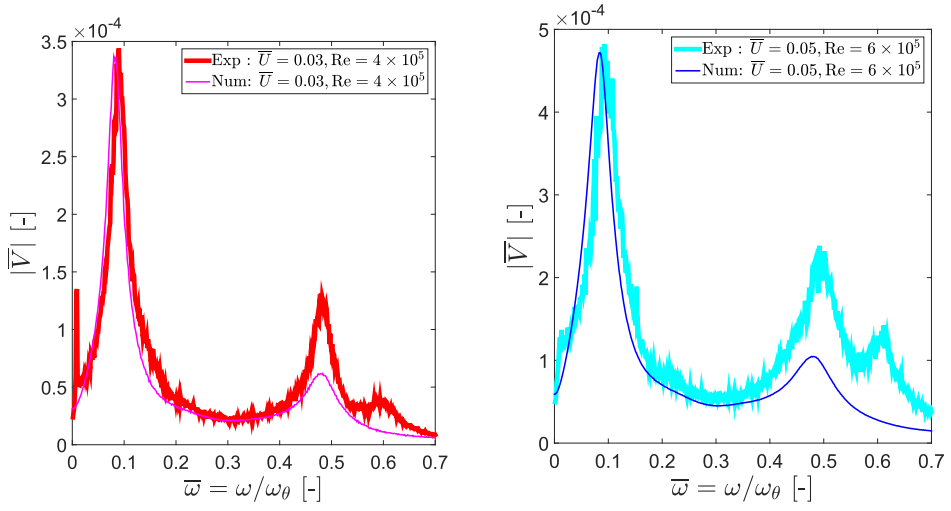


FIG. 11. Variation of the predicted and measured frequency spectra ($\overline{\omega} = \omega/\omega_\theta$) of the natural flow-induced tip bending and twisting velocity fluctuations, $\overline{V} = (\dot{h} + x\dot{\theta})/(\omega_\theta b)$, with varying reduced velocity ($\overline{U} = 0.03$ and 0.05, $\mathrm{Re} = 4 \times 10^5$ and 6×10^5) for a cantilevered flexible NACA0015 POM hydrofoil in water at $\alpha_o = 8°$, and $x = 0.04$ m. Note that the FFT window size was selected as $\overline{t} = 0$–2500.

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NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) at $\alpha_o = 8°$. Note that $\overline{V} = (\dot{h} + x\dot{\theta})/(\omega_\theta b)$, where $x = 0.04$ m is the distance from the reference point to the measurement point of the Laser Doppler Velocimetries (LDVs) measurement point near the free tip and the leading-edge on the pressure side of the foil, which corresponds to the lower green point in Fig. 1(d).

The results in Fig. 11 show that the numerical predictions of the bending and twisting frequency peaks obtained by using viscous FSI simulations agree reasonably well with the experimental measurements. The numerical model did not capture a third frequency peak that was observed in the experiment, because it corresponded to the second bending mode, but the current 2-DOF model is designed to capture the first bending and the first twisting modes only. Differences can be observed between the predicted and measured amplitudes, which are more sensitive to window size and initial conditions. Note that the inflow velocity was slowly increased in the experimental study for a given angle of attack, and then the angle of attack was systematically varied. In contrast, the numerical viscous FSI simulations was initialized with the steady-state solution at the particular $\overline{U}$ and α_o for the rigid case, and then foil motioned and viscous FSI responses were computed by solving Eq. (10). Nevertheless, the location of the frequency peaks should be the same, as the response is dominated by the natural frequencies in absence of vortex shedding due to flow separation.

In Secs. IV–VI, results are presented from the combined numerical and experimental studies to illustrate natural flow-induced vibrations of flexible hydrofoils as functions of the reduced velocity, angle of attack, and solid-to-fluid added mass ratio.

IV. INFLUENCE OF THE REDUCED VELOCITY

Detailed variations of the natural flow-induced vibration responses for a cantilevered flexible NACA0015 POM hydrofoil with varying inflow velocity (as reflected by the change in $\overline{U} = U/\omega_\theta b$, the reduced velocity) are shown in this section. For all cases, $\overline{U}$ was varied by fixing ω_θ and varying U; hence, Re was changed simultaneously with $\overline{U}$. Note that only fully turbulent flow is considered, i.e., $\mathrm{Re} \geq 4 \times 10^5$.

A. Time-histories of the flow-induced tip bending and twisting responses

Figure 12 shows the time-histories of the natural flow-induced spanwise tip bending and twisting deformations at the tip of a cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) with varying reduced velocity ($\overline{U} = 0.03$–0.08, $\mathrm{Re} = 4 \times 10^5$–$1 \times 10^6$) at $\alpha_o = 8°$. The results in Fig. 12 indicate that the amplitudes of bending and twisting deformations increase with $\overline{U}$, as $\alpha_o = 8°$ corresponds to pre-stall condition. It should be noted that $\alpha_{\mathrm{eff}} = \alpha_o - \theta$ and $\theta < 0$, so $\alpha_{\mathrm{eff}} > \alpha_o$ because flow induces a clockwise moment (with respect to the view given in Fig. 5) and rotation as the center of pressure (CP) is upstream of the elastic axis (EA) prior to stall. Stall occurs when $\alpha_{\mathrm{eff}} = 16°$–$18°$ for $\mathrm{Re} = 5 \times 10^5$–$1.2 \times 10^6$, based on the rigid NACA0015 results shown in Fig. 10(a).

B. Frequency spectra of the flow-induced tip bending and twisting responses

Figure 13 presents the predicted and measured variation of the frequency spectra via FFT method with the moving average function of the natural flow-induced tip bending and twisting velocity fluctuations with varying reduced velocity ($\overline{U} = 0.03$–0.1, $\mathrm{Re} = 4 \times 10^5$–$1.2 \times 10^6$) for a cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) at $\alpha_o = 8°$. Figure 13 shows that the viscous FSI model is able to capture the frequency peaks corresponding to the in-water natural bending and twisting frequencies. Note that the viscous FSI model only considers the first spanwise bending and first spanwise twisting degrees of freedom, and hence the third frequency peak corresponding to the second bending mode is not captured. In addition, experimental measurements were not available for $\overline{U} > 0.05$.

Table V lists the predicted and measured first in-water natural bending and twisting frequencies (f_h^*&f_θ^*) from Fig. 13. The results show that f_h^* and f_θ^* vary slightly with $\overline{U}$. Note that similar variation of the natural frequency and damping coefficient with velocity was also reported in Besch and Liu,[3–5] Blake and Maga,[6] Reese,[7] Seeley *et al.*,[8] Liaghat *et al.*,[9] and Yao *et al.*[10]

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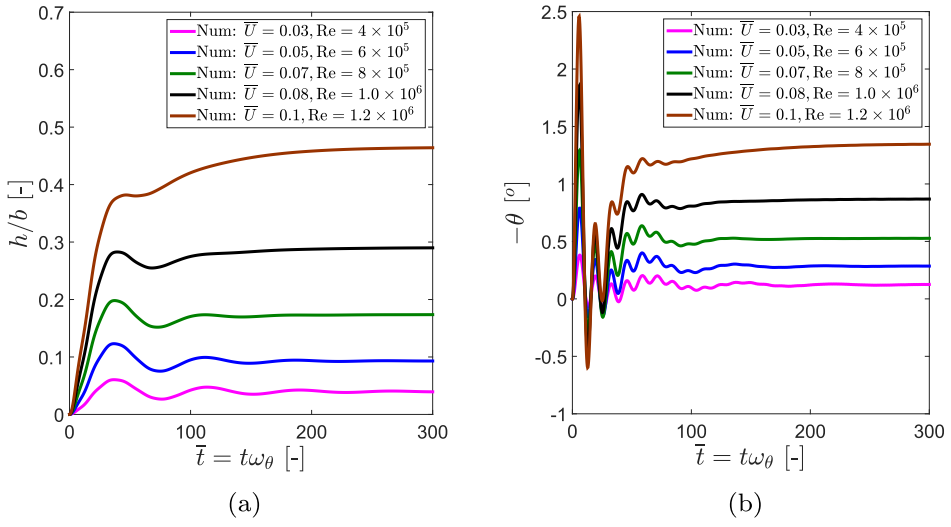


FIG. 12. Variation of the predicted time-histories of the natural flow-induced spanwise tip (a) bending deformations and (b) twisting deformations with varying reduced velocity at the tip of a cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) at $\alpha_o = 8°$.

C. Total loss factors of the flow-induced tip bending and twisting responses

Figure 14 shows a comparison of the total loss factors (η_T) obtained using the viscous FSI solver, the inviscid, *uncoupled mode* (UM), linear theory used by Blake and Maga[6] in Eqs. (22) and (23), the data from the current experiments for the NACA0015 POM hydrofoil, and the experimental measurements of Blake and Maga[6] for a stainless steel bullet shape hydrofoil and Reese[7] for an aluminum NACA66 hydrofoil. Viscous FSI simulations with different $\overline{U}$ were only conducted for the NACA0015 POM hydrofoil.

The assumed solid damping coefficients ($\zeta_{s,h} = \eta_{s,h}/2$ and $\zeta_{s,\theta} = \eta_{s,\theta}/2$) for the three different hydrofoils are shown in Table IV. Although the geometry and natural frequencies of the stainless steel and aluminum hydrofoils examined in the experimental studies by Blake and Maga[6] and by

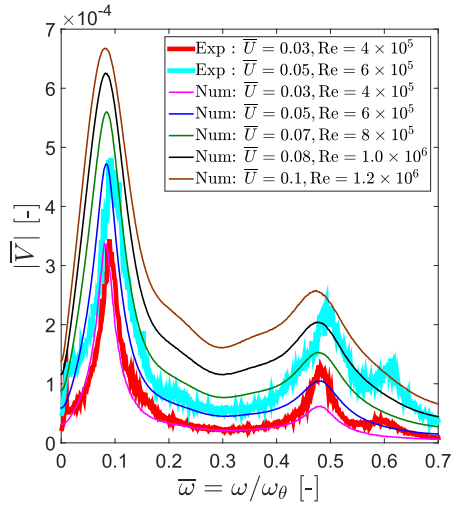


FIG. 13. Variation of the predicted and measured frequency spectra ($\overline{\omega} = \omega/\omega_\theta$) of the natural flow-induced tip bending and twisting velocity fluctuations ($\overline{V}$) with varying reduced velocity for a cantilevered flexible NACA0015 POM hydrofoil in water at $\alpha_o = 8°$. Note that the FFT window size was selected as $\overline{t} = 0$–2500, and experimental data were only available for $\overline{U} = 0.03$ and 0.05 at $\alpha_o = 8°$.

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TABLE V. Comparison of the predicted and measured first in-water natural bending and twisting frequencies (f_h^* and f_θ^*) at $\alpha_o = 8°$ from Fig. 13 for the cantilevered flexible NACA0015 POM hydrofoil in water. It should be noted that the numerical values with superscript "+" are computed using Theodorsen's[2] approach as they correspond to the still-water conditions.

$f_h = 81$ (Hz) $f_\theta = 390$ (Hz)		Still water Re = 0	$\overline{U} = 0.03$ Re = 4×10^5	$\overline{U} = 0.05$ Re = 6×10^5	$\overline{U} = 0.07$ Re = 8×10^5	$\overline{U} = 0.08$ Re = 1×10^6	$\overline{U} = 0.1$ Re = 1.2×10^6
f_h^* (Hz)	Exp	34	35	36	N/A	N/A	N/A
	Num.	33^+	31	33	33	32	30
f_θ^* (Hz)	Exp	184	187	192	N/A	N/A	N/A
	Num.	190^+	191	192	194	195	192

Reese,[7] respectively, are not the same as the current experimental and numerical model for the NACA0015 POM hydrofoil, the trends of the total loss factor with $\overline{U}$ for a given μ are similar. Since all cases shown in this work are significantly below the flutter and divergence speeds, the total loss factor tends to increase with increasing $\overline{U}$ and with decreasing μ (i.e., decreasing solid density for a given fluid), consistent with the trends as expected in Eqs. (22) and (23). Note that for a given foil with a fixed μ value, the total loss factor will first increase with $\overline{U}$, then decrease towards zero, as $\overline{U}$ further increases towards the critical flutter speed, as shown in Besch and Liu.[5] However, as shown in Chae, Akcabay, and Young,[43] for most rectangular hydrofoils with a span to chord ratio greater than one, static divergence (which occurs when the fluid disturbing moment is equal to or higher than the structural elastic restoring moment) tends to occur before flutter because of the

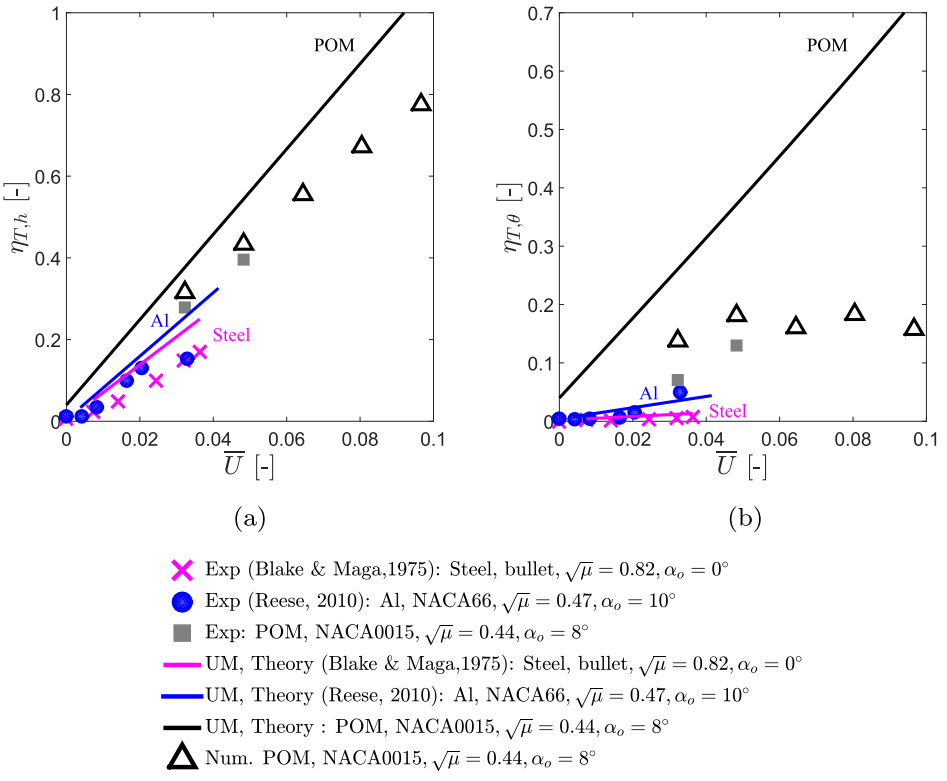


FIG. 14. Comparison of the measured and predicted (a) bending total loss factors ($\eta_{T,h}$) and (b) twisting total loss factors ($\eta_{T,\theta}$) with varying reduced velocity ($\overline{U}$) for the natural flow-induced spanwise bending and twisting deformations at the tip of a cantilevered flexible NACA0015 POM hydrofoil ($\sqrt{\mu} = 0.43$). Also shown are experimental measurements of a cantilevered stainless steel bullet shape hydrofoil ($\sqrt{\mu} = 0.82$) from Blake and Maga[6] and a cantilevered aluminum NACA66 hydrofoil ($\sqrt{\mu} = 0.47$) from Reese,[7] along with predictions of the inviscid, UM, linear theory by using Eqs. (22) and (23) according to Blake and Maga.[6]

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high fluid disturbing moment and high fluid damping. The static divergence speed, as estimated via $\overline{U_d} = \sqrt{\mu r_\theta^2/(1+2a)}$, corresponds to $\overline{U} = 0.28$ for the NACA0015 POM hydrofoil. However, it is dangerous to carry out experiments to such high speeds in a recirculating water tunnel. Hence, only $\overline{U} \leq 0.05$ was studied in the experiment. Numerical simulations were limited to $\overline{U} \leq 0.1$ to avoid excessive mesh deformations and issues associated with nonlinear interactions with the tunnel top and bottom no-slip boundaries. Note that the maximum tip bending and twisting deformations were $h/c = 0.46$ and $\theta = 2.4°$ at $\overline{U} = 0.1$.

The results in Fig. 14 show that the inviscid, UM, linear theory equation used in Blake and Maga[6] tends to over-predict the total loss factor for all three hydrofoils, and the over-prediction increases with decreasing μ and increasing $\overline{U}$. This is because the theory ignores the contribution of flow-induced bend-twist coupling of the hydrodynamic loads (i.e., the off-diagonal terms in $\tilde{\mathbf{C}}_f$ and $\tilde{\mathbf{K}}_f$ in Eqs. (19) and (20)), which should attain a greater importance with low μ and high $\overline{U}$. This over-prediction is not desirable since the unexpected lower damping can yield earlier material fatigue, a longer settling time, and increasing noise and vibration, as well as earlier susceptibility to flutter and divergence for $\sqrt{\mu} < 1$ (as observed in Chae, Akcabay, and Young[43]).

D. Wake structures

Figure 15 shows the snapshots of the predicted vorticity contours obtained using viscous FSI simulations at selected time instances for the cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) with $\overline{U} = 0.03$–0.1 ($\mathrm{Re} = 4 \times 10^5$–$1.2 \times 10^6$) at $\alpha_o = 8°$.

The results in Fig. 15 show that the unsteady vortex shedding can be observed during the transient responses before steady-state is reached (i.e., $\bar{t} = 0$–300). The results also show that as the inflow velocity increases, the shedding occurs earlier and the steady-state condition is approached faster with the increasing deformations, as already shown in Fig. 12.

V. INFLUENCE OF ANGLE OF ATTACK

The dynamic responses of flexible hydrofoils are expected to be sensitive to variations in the angle of attack (α_o) due to changes in the dynamic lift, CP, and wake patterns.[47] Detailed variations of the time and frequency responses of the natural flow-induced deformations with varying angle of attack are shown in this section.

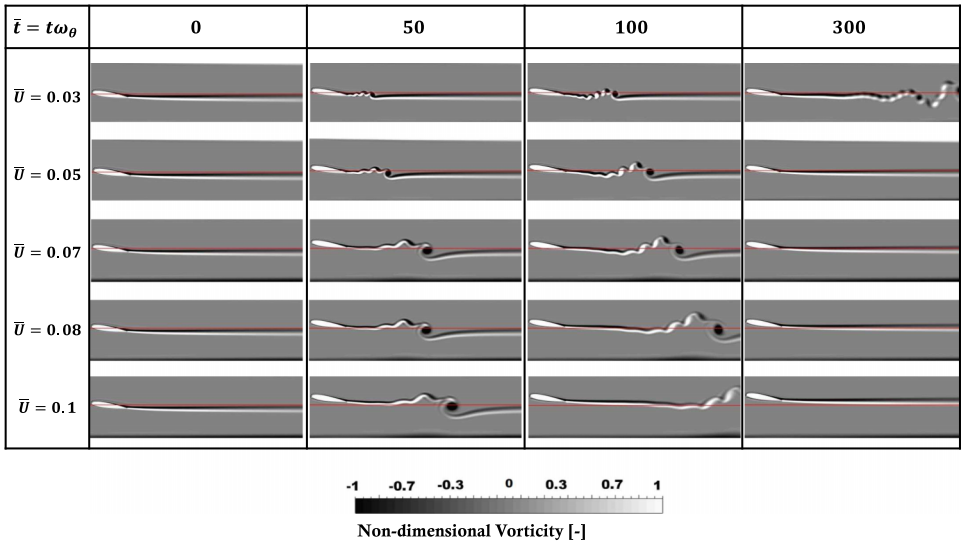


FIG. 15. Snapshots of the predicted non-dimensional vorticity contours ($\overline{w} = \omega b/U$) of the flow around the cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) at different time instances with $\overline{U} = 0.03$–0.1 ($\mathrm{Re} = 4 \times 10^5$–$1.2 \times 10^6$) at $\alpha_o = 8°$. The horizontal reference lines pass through the elastic axis of the undeformed hydrofoil.

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A. Time-histories of the flow-induced tip bending and twisting responses

Figure 16 shows the predicted time-histories of the natural flow-induced spanwise tip bending and twisting deformations of a cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) with varying angles of attack ($\alpha_o = 2°$–$20°$) at $\overline{U} = 0.05$ (Re $= 6 \times 10^5$). Before the foil stalls ($\alpha_{\text{eff}} < 16°$–$18°$), the amplitude of twisting deformation increases approximately linearly with the angle of attack because the elastic flow-induced deformations are linearly related to the lift force, which varies approximately linearly with the angle of attack until a trailing-edge vortex (TEV) develops at $\alpha_o \geq 15°$, as shown later in Figs. 19 and 20. After the foil stalls, $\alpha_{\text{eff}} > 16°$–$18°$, the mean lift drops, and the mean amplitudes of bending and twisting deformations also drop, along with a development of large amplitude oscillations due to dynamic vortex shedding.

B. Frequency spectra of the flow-induced tip bending and twisting responses

Figure 17 shows the variation of the predicted and measured frequency spectra of the natural flow-induced tip bending and twisting velocity fluctuations with different angles of attack for a cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) at $\overline{U} = 0.05$ (Re $= 6 \times 10^5$). The results show good comparisons of the frequency peaks between the predicted and measured responses at $\alpha_o = 2°$, $\alpha_o = 8°$, and $\alpha_o = 17°$.

Table VI compares the predicted and measured first in-water natural bending and twisting frequencies with angle of attack, which are also shown in Fig. 17. The results show that the in-water natural frequencies vary slightly with the angle of attack.

C. Total loss factors of the flow-induced tip bending and twisting responses

Figure 18 shows a comparison of the total loss factors (η_T) of the natural flow-induced spanwise tip bending and twisting deformations obtained using the current viscous FSI solver; the inviscid, *uncoupled mode* (UM), linear theory given by Blake and Maga[6] in Eqs. (22) and (23); as well as the experimental measurements of the current NACA0015 POM hydrofoil and Reese's NACA66 aluminum hydrofoil with varying angles of attack. Note that viscous FSI simulations are only shown for the NACA0015 POM hydrofoil ($\sqrt{\mu} = 0.43$).

The predicted viscous FSI results compare well with the experimental measurements at the same μ. The inviscid, UM, linear theory (as given in Eqs. (22) and (23)) overestimates the total

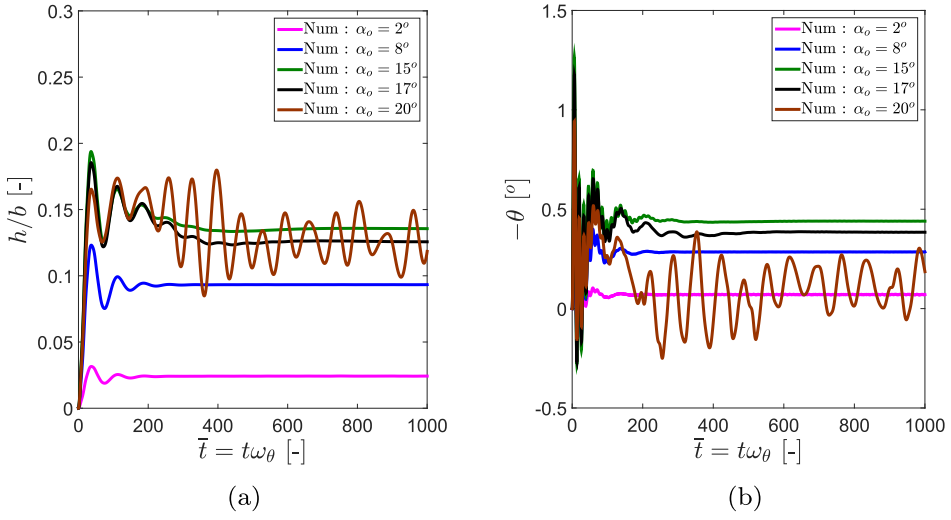


FIG. 16. Variation of the predicted time-histories of the natural flow-induced spanwise tip (a) bending deformations and (b) twisting deformations with varying angles of attack at the tip of a cantilevered flexible NACA0015 hydrofoil ($\sqrt{\mu} = 0.43$) in water at $\overline{U} = 0.05$ (Re $= 6 \times 10^5$).

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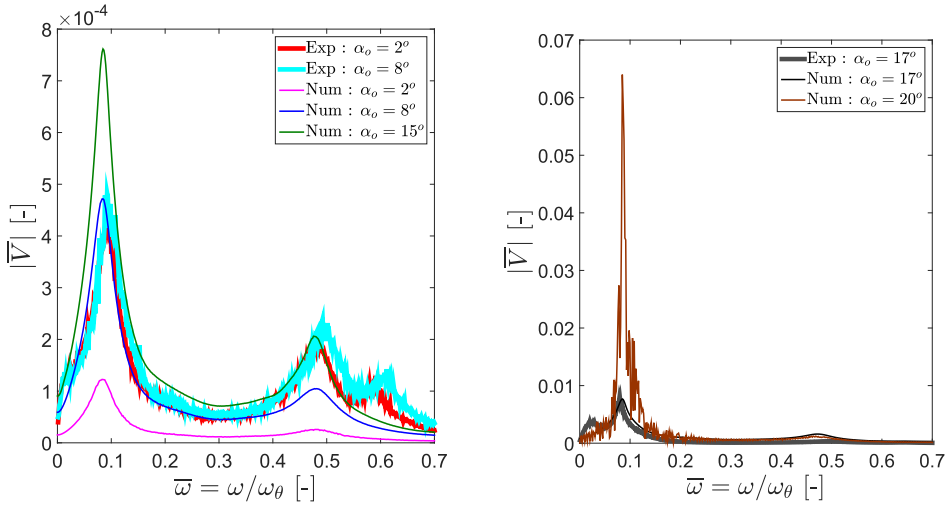


FIG. 17. Variation of the predicted and measured frequency spectra ($\overline{\omega} = \omega/\omega_\theta$) of the natural flow-induced tip bending and twisting velocity fluctuations with varying angles of attack for a cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) at $\overline{U} = 0.05$ (Re = 6×10^5). Note that the FFT window size is $\bar{t} = 0$–2500, and experimental data were only available for $\alpha_o = 2°$, 8°, and 17°.

loss factor for both POM and Al hydrofoils. Note that the inviscid theory also cannot predict the variation of the total loss factor with the angle of attack, which are caused by viscous effects such as flow separation and unsteady vortex shedding at high angles of attack or transitional flows. The over-prediction of η_T by the inviscid, UM, linear theory is more severe for the lighter NACA0015 POM hydrofoil than the heavier NACA66 Al hydrofoil studied by Reese.[7] At a post-stall angle of attack of $\alpha_o = 20°$ with $\overline{U} = 0.05$ (Re = 6×10^5), the natural flow-induced spanwise bending and twisting deformations get into a bending limit-cycle behavior, as shown in Fig. 16, and hence the bending loss factor becomes very low and the amplitude of the flow-induced vibrations at the bending natural frequency becomes very high, as shown in Figs. 17 and 18.

D. Wake structures

Figure 19 shows the snapshots of the predicted vorticity contours of viscous FSI simulations at selected time instances on the cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) at $\alpha_o = 2°$–20° and $\overline{U} = 0.05$ (Re = 6×10^5). Figure 20 shows the streamlines of the flow around the cantilevered rigid and flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) at $t' = tU/b = 70$, which is the steady-state condition.

The results in Fig. 19 show that the vortex shedding can be observed even for the $\alpha_o = 2°$ case during the transient phase prior to reaching static equilibrium (i.e., $\bar{t} = 0$–300). As shown in Fig. 20, at $\alpha_o \geq 15°$, a larger trailing-edge vortex (TEV) is observed on the flexible foil than the rigid foil due to increases in the effective angle of attack caused by the clockwise flow-induced twisting

TABLE VI. Comparison of the predicted and measured first in-water natural bending and twisting frequencies for a cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) for $\overline{U} = 0.05$ (Re = 6×10^5) from Fig. 17 (f_h and f_θ are the in-air natural frequencies, and f_h^* and f_θ^* are the in-water natural frequencies).

$f_h = 81$ (Hz), $f_\theta = 390$ (Hz)		$\alpha_o = 2°$	$\alpha_o = 8°$	$\alpha_o = 15°$	$\alpha_o = 17°$	$\alpha_o = 20°$
f_h^* (Hz)	Exp	36	36	N/A	31	N/A
	Num.	32	33	33	33	34
f_θ^* (Hz)	Exp	187	192	N/A	192	N/A
	Num.	193	192	189	188	189

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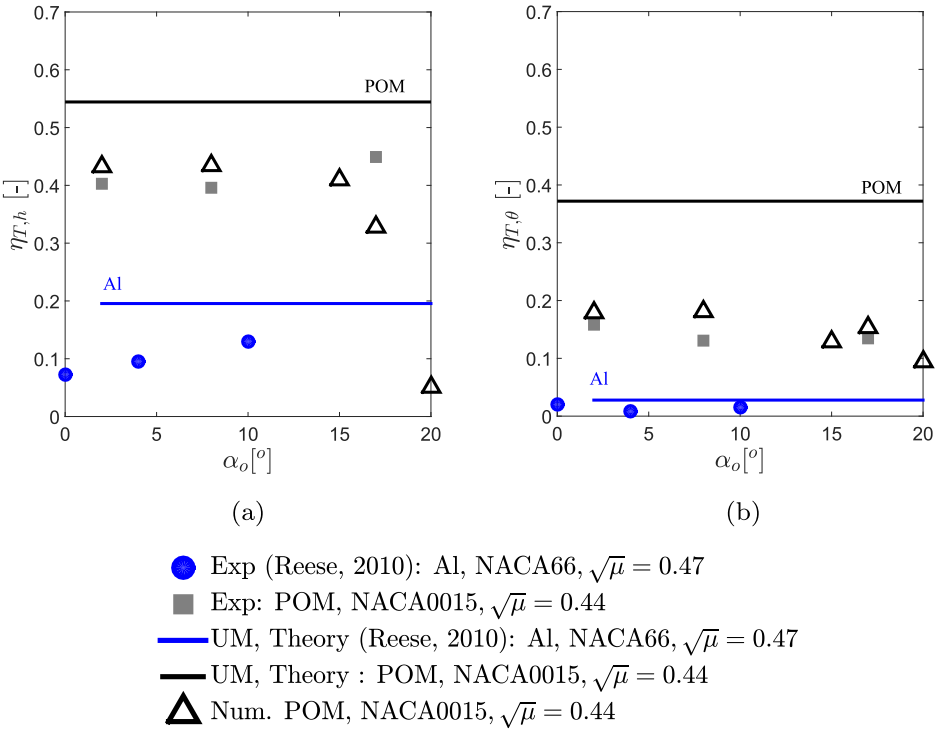


FIG. 18. Comparison of the measured and predicted (a) bending total loss factors ($\eta_{T,h}$) and (b) twisting total loss factors ($\eta_{T,\theta}$) with varying angle of attack (α_o) for the natural flow-induced spanwise bending and twisting deformations at the tip of a cantilevered flexible NACA0015 POM hydrofoil ($\sqrt{\mu} = 0.43$) at $\overline{U} = 0.05$ ($\mathrm{Re} = 6 \times 10^5$). Also shown are experimental measurements of an aluminum NACA66 hydrofoil in water ($\sqrt{\mu} = 0.47$) from Reese,[7] as well as predictions of the inviscid, UM, linear theory by using Eqs. (22) and (23). Note that viscous FSI simulations are only shown for the NACA0015 POM hydrofoil.

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motion, which leads to slightly earlier stall of the flexible hydrofoil. Particularly, at $\alpha_o = 20°$, the flow is fully separated for both rigid and flexible hydrofoils, as shown in Figs. 19 and 20. It should be noted that for the case shown in Fig. 19, the vortex shedding frequency of the flexible hydrofoil is close to the first in-water natural bending frequency, i.e., near lock-in, at $\alpha_o = 20°$ and $\overline{U} = 0.05$. At this point, large-scale fluctuations in loads and deformations (Figs. 16 and 19) are observed, resulting in high peak vibration amplitude at the natural bending frequency (Fig. 17(b)), and a low bending damping coefficient (Fig. 18).

VI. THE EFFECT OF THE SOLID-TO-FLUID ADDED MASS RATIO (μ)

Detailed variations of the natural flow-induced vibration responses for the cantilevered flexible NACA0015 hydrofoils with varying solid-to-fluid added mass ratios (μ) are shown in this section. The solid-to-fluid added mass ratio, $\mu = \tilde{m}/(\pi \rho_f b^2 s)$, is varied by changing the solid density e.g., polyacetate (POM), aluminum (Al), and stainless steel (steel), for a given fluid density and semi-chord length.

A. Time-histories of the flow-induced tip bending and twisting responses

Figure 21 shows the sample time-histories of the natural flow-induced spanwise tip bending and twisting deformations of the cantilevered flexible NACA0015 hydrofoils with varying solid-to-fluid added mass ratios at $\alpha_o = 8°$ and $\overline{U} = 0.05$ ($\mathrm{Re} = 6 \times 10^5$–$2.1 \times 10^6$). Note that in order to achieve $\overline{U} = 0.05$ for the hydrofoils made of different materials, the inflow velocity was varied to account for the material dependence of the in-air natural twisting frequency ($\omega_\theta = 2\pi f_\theta$), as shown in Table IV.

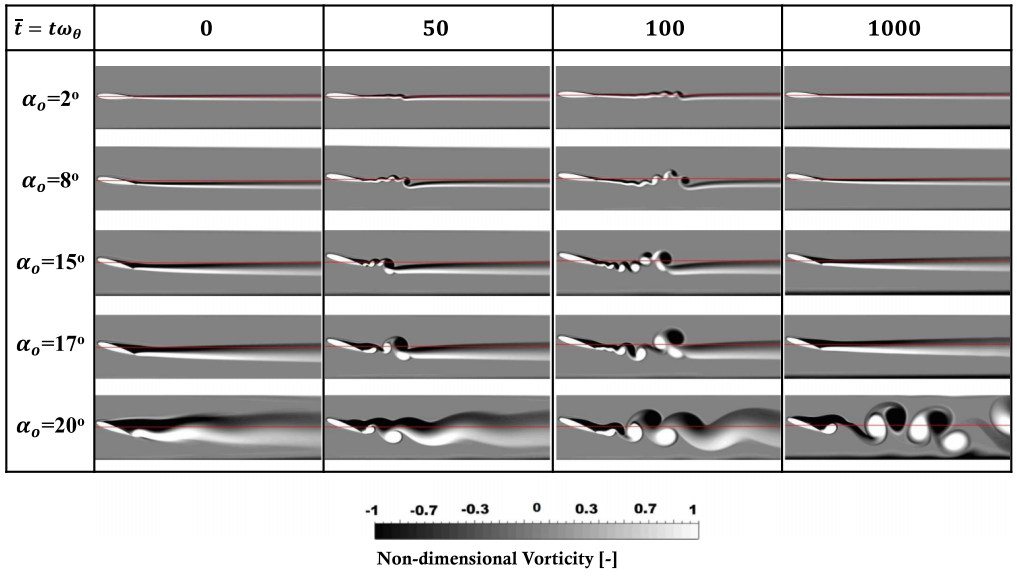


FIG. 19. Snapshots of the predicted non-dimensional vorticity contours ($\overline{w} = \omega b/U$) of the flow around the cantilevered flexible NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) at different time instances with $\alpha_o = 2°$–$20°$ and $\overline{U} = 0.05$ ($\mathrm{Re} = 6 \times 10^5$). The horizontal reference lines pass through the elastic axis of the undeformed hydrofoil.

The results in Fig. 21 show that the amplitudes of the natural flow-induced spanwise tip bending and twisting deformations increase, and damp out quicker, as the solid-to-fluid added mass ratio decreases (i.e., material becomes lighter for a given fluid) due to the increased fluid load compared to the solid load, and due to the reduction in solid stiffness, as indicated by the changing E_s values for the three different materials in Table IV.

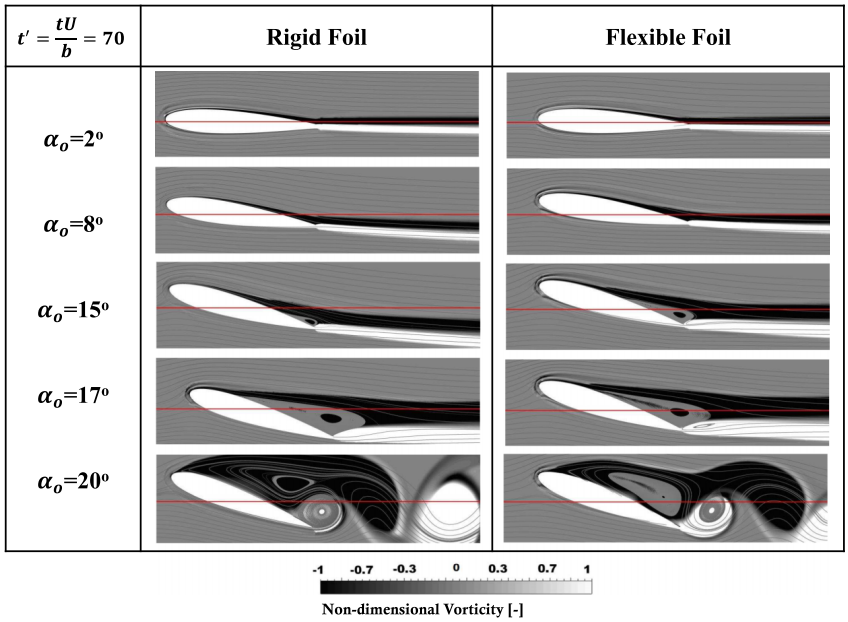


FIG. 20. Snapshots of the streamlines of the flow around the cantilevered flexible and rigid NACA0015 POM hydrofoil in water ($\sqrt{\mu} = 0.43$) at $\alpha_o = 2°$–$20°$, $\mathrm{Re} = 6 \times 10^5$, and $t' = tU/b = 70$. The horizontal reference lines pass through the elastic axis of the undeformed hydrofoil. Notice that $\omega_\theta = 2\pi f_\theta$ for the rigid foil is theoretically infinite. To allow consistent comparisons with the flexible foil results, a new non-dimensional time, t', is used.

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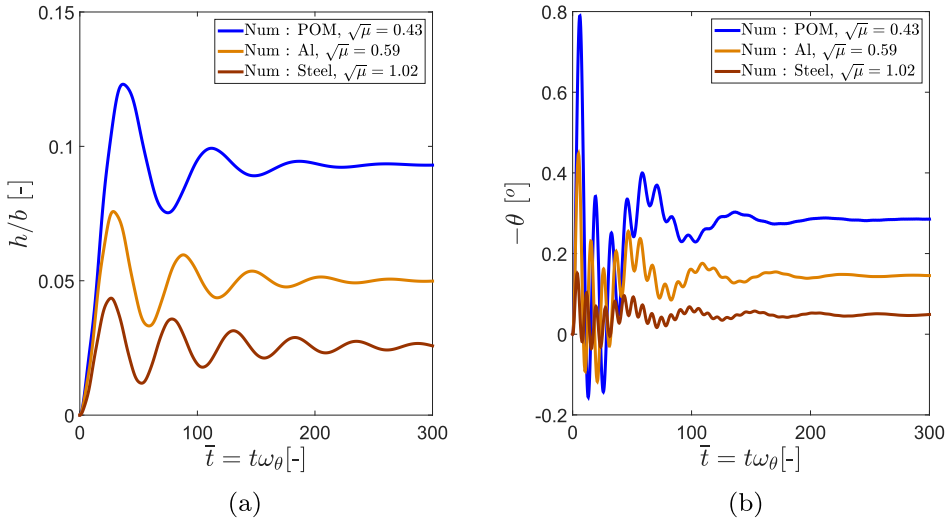


FIG. 21. Variation of the predicted time-histories of the natural flow-induced spanwise tip (a) bending deformations and (b) twisting deformations with varying solid-to-fluid added mass ratios ($\sqrt{\mu} = 0.43, 0.59, 1.02$) at the tip of the cantilevered flexible NACA0015 hydrofoils in water at $\alpha_o = 8°$ and $\overline{U} = 0.05$ (Re = 6×10^5–2.1×10^6).

B. Frequency spectra of the flow-induced tip bending and twisting responses

Figure 22 presents the predicted and measured frequency spectra and non-dimensional natural bending ($\overline{\omega_h}$) and twisting ($\overline{\omega_\theta}$) frequencies of the natural flow-induced tip bending and twisting velocity fluctuations with varying solid-to-fluid added mass ratios ($\sqrt{\mu}$ = 0.43–1.02) for the cantilevered flexible NACA0015 hydrofoils at α_o = 8° and $\overline{U}$ = 0.05. The results show that in-water natural bending and twisting frequencies (f_h^* and f_θ^*) decrease rapidly with reductions in the solid-to-fluid added mass ratio, i.e., as the material becomes lighter, due to the greater relative importance of fluid added mass. The results in Fig. 22(b) show that good agreement is observed

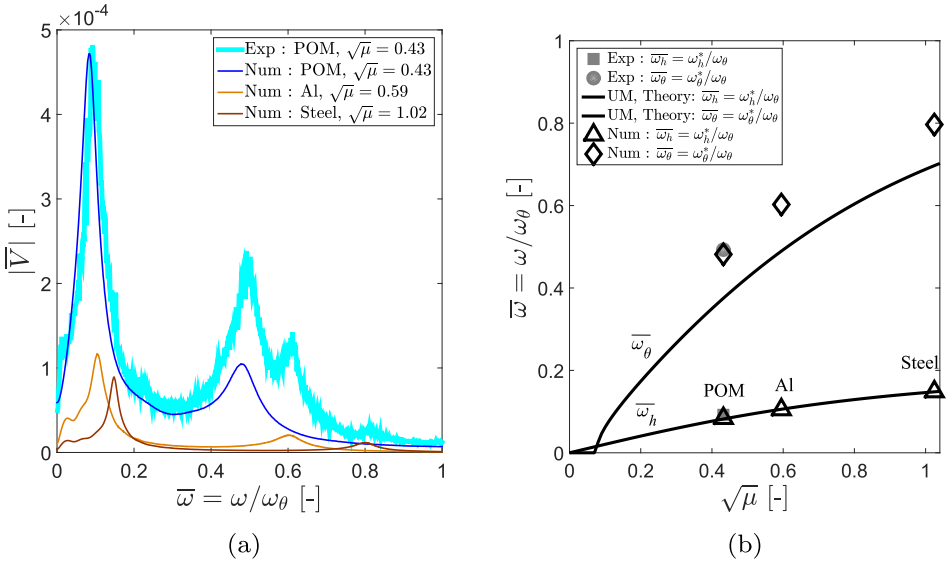


FIG. 22. Variation of the predicted and measured (a) frequency spectra ($\overline{\omega} = \omega/\omega_\theta$) and (b) non-dimensional natural bending ($\overline{\omega_h}$) and twisting ($\overline{\omega_\theta}$) frequencies of the natural flow-induced tip bending and twisting velocity fluctuations with varying solid-to-fluid added mass ratios ($\sqrt{\mu} = 0.43, 0.59, 1.02$) on the cantilevered flexible NACA0015 hydrofoils in water at $\alpha_o = 8°$ and $\overline{U} = 0.05$ (Re = 6×10^5–2.1×10^6). The FFT window size was selected as $\overline{t}$ = 0–2500.

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between the bending natural frequency obtained using the inviscid, UM, linear theory, experimental and numerical results; however, the inviscid, UM, linear theory under-predicted the twisting natural frequency.

C. Total loss factors of the flow-induced tip bending and twisting responses

Figure 23 shows the total loss factors ($\eta_{T,h}$ and $\eta_{T,\theta}$) with varying solid-to-fluid added mass ratios for the cantilevered flexible NACA0015 hydrofoils at $\alpha_o = 8°$ and $\overline{U} = 0.05$. The results show that the inviscid, UM, linear theory given in Eqs. (22) and (23) over-estimates the total loss factors, particularly for the twisting deformations, compared with the viscous and coupled FSI predictions and experimental measurement. This increased loss factor (i.e., damping) leads to non-harmonic flow-induced spanwise bending and twisting deformations with rapidly decaying oscillations, as shown in Fig. 21. It should be noted that the rapid increase of the total loss factor with decreasing μ is the reason why flutter, which occurs when total effective damping is zero, is usually not a great concern for hydrofoils compared to airfoils. Instead, hydrofoils are typically governed by static divergence instability due to the high fluid disturbing moment, as noted in Chae, Akcabay, and Young.[43]

D. Wake structures

Figure 24 shows the snapshots of the predicted vorticity contours of viscous FSI simulations at selected time instances on the cantilevered flexible NACA0015 hydrofoils in water with varying solid-to-fluid added mass ratios ($\sqrt{\mu} = 0.43$–1.02) at $\overline{U} = 0.05$ and $\alpha_o = 8°$. The results in Fig. 24 show that the unsteady vortex shedding can be observed during the transient responses (i.e., $\bar{t} = 0$–300), and the steady-state condition is approached faster for the low solid-to-fluid added mass ratio case (i.e., $\sqrt{\mu} = 0.43$, POM hydrofoil), as already shown in Fig. 21.

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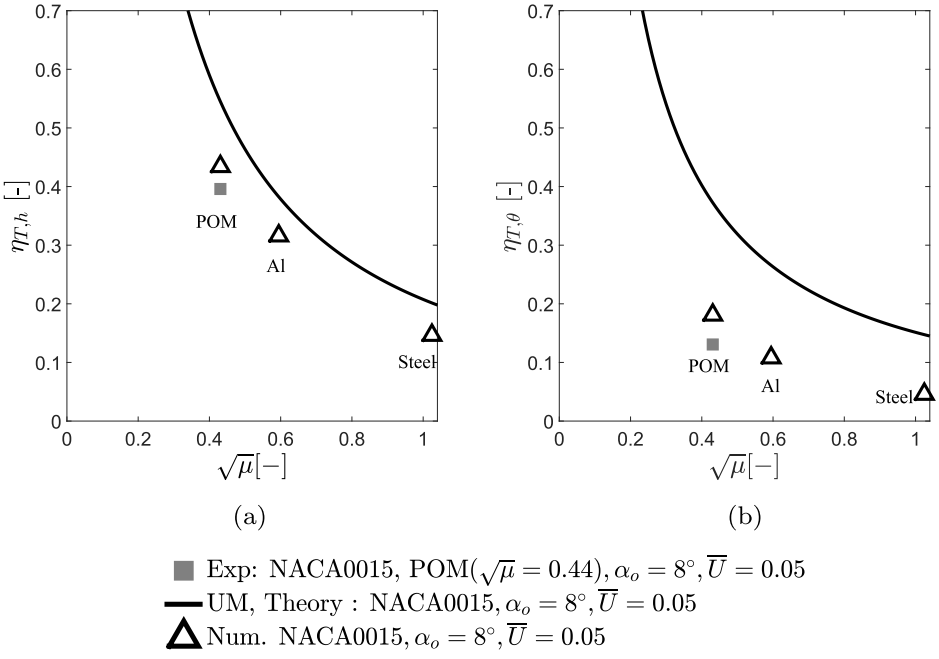


FIG. 23. Comparison of the measured and predicted (a) bending total loss factors ($\eta_{T,h}$) and (b) twisting total loss factors ($\eta_{T,\theta}$) with varying solid-to-fluid added mass ratios (μ) for the natural flow-induced spanwise bending and twisting deformations at the tip of a cantilevered flexible NACA0015 hydrofoils in water at $\alpha_o = 8°$ and $\overline{U} = 0.05$ ($\mathrm{Re} = 6\times10^5$–$2.1\times10^6$). Note that the inviscid, UM, linear theory uses Eqs. (22) and (23).

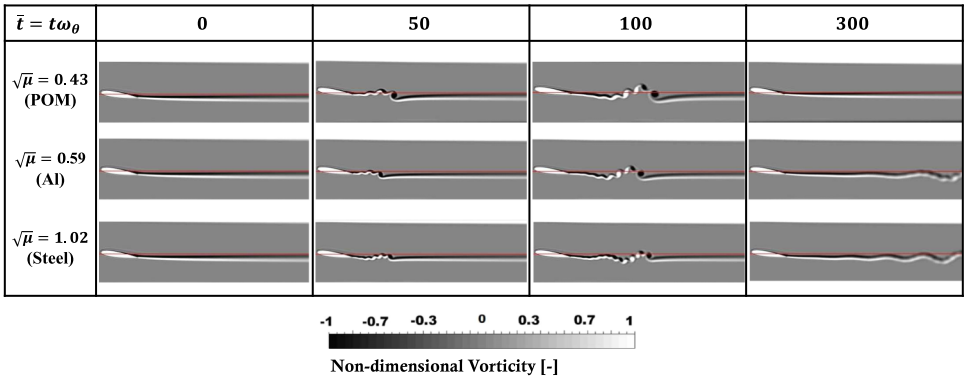


FIG. 24. Snapshots of the predicted non-dimensional vorticity contours ($\overline{w} = \omega b/U$) of the flow around the cantilevered flexible NACA0015 hydrofoils in water with varying solid-to-fluid added mass ratios ($\sqrt{\mu} = 0.43$–1.2) at $\overline{U} = 0.05$ and $\alpha_o = 8°$. The horizontal reference lines pass through the elastic axis of the undeformed hydrofoil.

VII. SUMMARY

The natural flow-induced vibrations of rectangular, cantilevered flexible hydrofoils with spanwise bending and twisting flexibilities were studied via combined numerical and experimental studies by varying the inflow velocity, angle of attack, and solid-to-fluid added mass ratio. The viscous fluid-structure interaction (FSI) response of the hydrofoil was modeled by using the loose hybrid coupled (LHC) method presented in Young, Chae, and Akcabay,[42] Chae, Akcabay, and Young,[43] and Akcabay *et al.*[33] to couple an unsteady Reynolds-averaged Navier-Stokes (URANS) fluid solver with $k-\omega$ shear stress transport ($k-\omega$ SST) turbulence model and a two-degree-of-freedom (2-DOF) generalized equation of motion model (EOM) representing the spanwise tip bending and twisting deformations of a cantilevered hydrofoil. The flow was assumed to be fully turbulent over the hydrofoil and effectively two-dimensional (2D) because of the high lift retention provided by the small gap between the foil tip and tunnel wall, and limited twisting deformation.

The simulations were performed on the NACA0015 hydrofoils in water made of the polyacetate (POM), aluminum (Al), and stainless steel (steel) materials with a chord length of 0.1 m and a span length of 0.191 m. The ranges of the reduced velocities ($\overline{U} = 0.03$–0.1 i.e., Re = 4×10^5–2.1×10^6), the angles of attack ($\alpha_o = 2°$–$20°$), and the solid-to-fluid added mass ratios ($\sqrt{\mu} = 0.43$–1.02) were chosen to obtain the viscous FSI responses. The model was validated by comparing the numerical predictions with experimental measurements conducted inside a cavitation tunnel at the French Naval Academy Research Institute (IRENav). Comparisons between the numerical and experimental results with prior published results of other flexible hydrofoils undergoing natural flow-induced vibrations, along with inviscid, uncoupled mode (UM), linear theory predictions were also shown. The results indicated the following.

- Numerical viscous FSI predictions of the first in-water natural bending and twisting frequencies, as well as total loss factors, compare well with available experimental measurements conducted at the IRENav.
- The foil vibrations are found to be dominated by the natural frequencies in absence of large scale vortex shedding due to flow separation.
- For lightweight hydrofoils, as the solid-to-fluid added mass ratio decreases, the natural frequencies reduce rapidly while the total loss factors increase rapidly. Furthermore, the natural frequencies and total loss factors vary with velocity, which is consistent with results reported in Besch and Liu,[3–5] Blake and Maga,[6] and Reese.[7] Consequently, the flow-induced vibrations will rapidly damp out while the fluid disturbing forces/moments increase with decreasing μ, which is why the governing instability mode will transition from flutter for $\sqrt{\mu} \geq 2$ to dynamic divergence for $1 \leq \sqrt{\mu} < 2$ to static divergence for $\sqrt{\mu} < 1$, as already shown in Chae, Akcabay, and Young.[43]

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- The inviscid, UM, linear theory provided in Blake and Maga[6] tends to over-predict the total loss factors for lightweight lifting bodies operating in dense fluids ($\sqrt{\mu} < 1$), particularly for the twisting vibrations. The over-prediction increases with increasing $\overline{U}$ and with decreasing μ. The over-prediction is mostly caused by neglecting of the flow-induced bend-twist coupling terms and viscous FSI effects, which are more significant for cases with low μ and high $\overline{U}$.

VIII. FUTURE WORK

For future work, additional experimental validation studies of the vortex shedding frequencies, natural frequencies, and total loss factors for both rigid and flexible hydrofoils with higher angles of attack and higher inflow velocities are needed. In particular, additional numerical and experimental studies are needed to investigate 3D effects and dynamic FSI responses at post-stall condition, where the unsteady vortex shedding frequency may lock in to one of the foil natural frequencies, and 3D flow effects caused by foil spanwise/chordwise deformations, gap flow, and tip vortex are not negligible. High fidelity 3D large-eddy simulation (LES), detached eddy simulation (DES), or even direct numerical simulation (DNS)[46] coupled with 3D flexible model is also needed to better quantify the dynamic FSI response and stability of lightweight flexible lifting bodies, particularly on the role of flow transition, massive flow separation, gap flow, and tip vortex dynamics on the foil response and stability.

ACKNOWLEDGMENTS

The authors gratefully acknowledge Dr. Ki-Han Kim and Ms. Kelly Cooper, program managers, and the Office of Naval Research (ONR), for their financial support through Grant Nos. N00104-13-1-0383 and N00014-11-1-0833, as well as ONR Global and Dr. Woei-Min Lin (program manager) through Grant No. N62909-12-1-7076.

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